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Polytope Fraud Theory

Dongshuai Zhao [a], Zhongli Wang [b], Florian Schweizer-Gamborino [c], Didier Sornette [d,*]

[a] KNK Capital, Melbourne, Australia
[b] Department of Business Administration and Economics, Universität Bielefeld, Germany and Department of Economics & Finance, Università Cattolica del Sacro Cuore, Italy
[c] PricewaterhouseCoopers AG (PwC), Switzerland
[d] Swiss Finance Institute, c/o University of Geneva, Geneva, Switzerland, and Institute of Risk Analysis, Prediction and Management (Risks-X), Southern University of Science and Technology (SUSTech), Shenzhen, China

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ABSTRACT

Polytope Fraud Theory (PFT) extends the existing triangle and diamond theories of accounting fraud with ten abnormal financial practice alarms that a fraudulent firm might trigger. These warning signals are identified through evaluation of the shorting behavior of sophisticated activist short sellers, which are used to train several supervised machine learning methods in detecting financial statement fraud using published accounting data. Our contributions include a systematic manual collection and labeling of companies that are shorted by professional activist short sellers. We also combine well-known asset pricing factors with accounting red flags in financial features selections. Using 80 % of the data for training and the remaining 20 % for out-of-sample test and performance assessment, we find that the best method is XGBoost, with a Recall of 72 % and F1-score of 82 %. Other methods have relatively lower performance, demonstrating the robustness of our results. This shows that the sophisticated activist short sellers, from whom the algorithms are learning, have excellent accounting insights, tremendous forensic analytical knowledge, and sharp business acumen. Our feature importance analysis indicates that potential short-selling targets share many similar financial characteristics, such as bankruptcy or financial distress risk, clustering in some industries, inconsistency of profitability, high accrual, and unreasonable business operations. Our results imply the possible automation of advanced financial statement analysis, which can both improve auditing processes and effectively enhance investment performance. Finally, we propose the Unified Investor Protection Framework, summarizing and categorizing investor-protection related theories from the macro-level to the micro-level.

The old wheel turns, and the same spoke comes up. It's all been done before, and will be again.
——— *Sherlock Holmes*

1. Introduction

Financial statement fraud has been, and remains, a significant concern worldwide for decades. It affects various industries, countries, and communities. In the U.S., the total cost[1] of financial statement manipulation has been estimated at roughly 572 billion dollars per year (Perols, 2011). In addition to direct costs, this fraud also negatively impacts suppliers, clients, employees, creditors, and investors, because inaccurate financial information interferes with their decision-making processes. Accounting fraud not only reduces financial market efficiency, but also decreases market trust, since financial accounting has played a critical role in modern economic activities. Investors rely on financial statements to make value-enhancing decisions and allocate their capital to the right places; bankers access them to decide whether the bank should grant credit; suppliers use them to evaluate whether clients are reliable; and regulators, tax authorities, customers, and even the most basic employees rely on financial statements to obtain

* Corresponding author.
E-mail address: dsornette@ethz.ch (D. Sornette).

[1] Perols (2011) estimated the total cost of financial statement fraud by assuming that the mean cost per case is equal to the median cost per fraud category. He used the number of cases times the mean cost per case to derive the total cost. The median cost as well as the number of cases are provided by the Association of Certified Fraud Examiners (Association of Certified Fraud Examiners (ACFE), 2008).

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information about businesses. Thus, it is clear that financial statements build bridges for corporate insiders and outsiders to communicate a company's financial soundness, credit risks, business strategies, and future perspectives. However, despite strong national and international monetary regulations and oversight accounting, corporate fraud scandals such as Enron, Lucent, WorldCom, Toshiba, Wirecard, and Evergrande continue to rock the financial world with astonishing regularity.

The financial statement fraud is usually defined as making falsified financial accounting statements by overstating balance sheet items, revenues, and net earnings, misappropriating taxes, or understating financial liabilities, expenses, or losses. Young and Cohen (2014) distinguish between deliberate fraud and financial statement errors, attributing financial statement errors to (1) limits to measuring technology (i.e., the difficulty of evaluating the technology "know-how" in dollar value), (2) randomness,[2] and (3) the subjective "accounting choices" of management (i.e., managers intentionally delaying or advancing reporting revenue to tactically smooth earnings). Young (2020) proposes that:

$$\text{Financial accounting} = \text{Economic reality} + \text{Error}.$$

In a comparison, accounting fraud is the intentional, material, perennial, and systematic misstatement of accounting reports, which might have serious consequences. Financial statement fraud is mainly related to inflating accounting numbers in income statements and balance sheets. We therefore borrow from the terminology of financial bubbles, which was used to describe the excess market prices above fundamental value, and we categorize the existence of two major types of fraud bubbles[3]: "income statement fraud bubbles" and "balance sheet fraud bubbles". The above equation can therefore be rewritten as:

$$\text{Financial accounting} = \text{Economic reality} + \text{Fraud bubbles} + \text{Error}.$$

"Income statement fraud bubble" refers to situations in which managers intentionally inflate the revenue or boost the net incomes[4] during some accounting periods. In contrast, a "balance sheet fraud bubble" is the situation in which managers intentionally manipulate corporate leverage or increase net assets by inflating the assets' value, hiding unfavorable debt,[5] or both.

An income statement fraud bubble accumulates over time and managers may want to hide the inflated net incomes within other items in the balance sheet, or they may boost expenses and costs to eliminate the inflated revenue over time. To hide inflated net incomes, they can either choose some items within the balance sheet to park the inflated income, or quietly spend the fake income through raw material purchases, Capex (capital expenditure) investments such as PP&E (property, plant, and equipment), or merger acquisitions (transforming the fake income into goodwill or intangible assets), depending on specific situations and difficulties. Nevertheless, if a company has accumulated too much inflated income on its balance sheet, the asset column on its balance sheet will become too bloated, and its financial ratios will be dragged down. After a certain period, such companies have to take a "big bath" to clean the balance sheet to "squeeze out the bubbles".

Financial fraud detection is very challenging, since it is:

- hard to confirm: there is no clear and exact accounting rule to decide whether a company is committing financial statement fraud. In practice, auditors, based on experience, can accept small, non-material deviations between reported and tracked accounts to allow for accounting error. Additionally, auditors have some leeway to allow for different accounting practices, as some companies might use conservative methods to smooth earnings, while companies which commit financial fraud tend to have bigger differences (for example, an activist short seller, Muddy Water Research, focuses on companies whose reported numbers deviate by more than 50 % from what would commonly be accepted as accounting reality).
- hard to detect: the traditional approach for detecting financial statement fraud is based on auditing procedures, which are time-consuming and sometimes unrealistic.[6] First, auditors usually lack the resources to apply sophisticated knowledge to identify financial fraud or prevent accounting malpractice. Second, due to the infrequent nature of such cases, in conjunction with the incentive structure of auditor remuneration, most auditors are incentivized to not notice accounting distortions. Third, corporate financial experts such as Chief Financial Officers (CFOs), financial managers, and accountants could intentionally use their deep industry expertise to deliberately deceive internal or external auditors. Finally, corporate insiders and auditors might even conspire to manipulate accounting numbers due to principal-agent problems and lack of self-regulation (for example, the Enron scandal).

Investment firms and financial institutions[7] can be seen as financial statement data collectors, users, and processors, and their tasks require some degree of intensive and specialized data analysis. Since the quantity of data available for analysis has grown tremendously, improving the accuracy and consistency of data analysis and interpretation poses a formidable challenge to financial statement users. Generally, institutions can expand their workforce to handle more financial data. Still, with the limited number of appropriate financial experts and the complexity of the data, it is quite challenging to scale up effective and accurate business decisions. Thus, traditional auditing and financial analysis procedures are far from meeting the demands of investors and creditors, suggesting the need for additional advanced, automatic financial statement analysis tools or techniques.

Recent improvements in computational intelligence (CI)-based techniques present potential solutions. Applying statistical methods or machine learning algorithms in financial statement analysis is a logical development. Much of the work comprises classification problems, such as fraud detection, bankruptcy prediction, crediting-rating evaluation, etc. Machine learning algorithms – which allow machines to uncover patterns without explicit specification of what to look for – are widely used in voice recognition (e.g., Siri), image recognition (e.g., self-driving cars), ranking systems (e.g., Google PageRank), and recommendation systems (e.g., Amazon's product recommendations). In accounting and finance, machine learning algorithms have successfully been applied to risk assessments, creditworthiness checks, lending evaluations, fraud detection auditing, automation, etc.

Machine learning algorithms present three crucial advantages compared with traditional manual detection procedures:

1. They can uncover nonlinear or hidden relationships and complex patterns.

[2] Randomness comes from manager estimates. For example, when a manager is estimating the cost of restructuring, it is impossible to predict exact cost outcomes before the restructuring occurs. The difference between actual cost and estimated cost is a typical source of randomness.

[3] We define a financial fraud bubble as a situation in which no reasonable fundamental reality can justify the (inflated) accounting number.

[4] Managers can artificially boost the net earnings by increasing revenue-related items, decreasing costs and expenses incurred, or both.

[5] For example, managers can artificially increase the assets' valuation, or conduct fake impairment tests on assets, or record less amortization cost of intangible assets.

[6] With limited time, it is almost impossible for auditors to correctly evaluate (1) the impairment rules of high-tech equipment; (2) the fair value of assets located among different countries or in some unknown rural areas; and (3) the market value of fishery resources and forestry resources, etc. in a very short period before the annual report is released.

[7] Such as loan associations, credit unions, mortgage companies, and commercial banks.

2. They show distinguishing-features extraction capability of large datasets without requiring specific knowledge of the input variables.
3. The algorithms are more scalable, standardized, and objective.

However, there are also some limitations[8] of using machine learning algorithms:

1. The difficulty of dealing with missing data or inadequate datasets.
2. Proneness to overfitting.
3. It can be challenging to interpret the model, because the result tends to be more quantitative, and the relationships among elements of the model are complex and multidimensional, leading to a "black box" situation.
4. Further investigation may be needed when the underlying situation changes substantially since the assumptions of the training regime might have been completely changed by the boundary conditions.
5. The quality of the output is highly dependent on the quality of the input data, and in many cases much of the effort is spent on bringing the data into a machine-readable, cleaned-up format.

Activist short sellers use their advanced accounting and analytical research skills, and specific information processing abilities to find potential short-selling targets. They short not only fraudulent companies, but also overvalued companies; sometimes short sellers hold short positions because they simply follow a momentum-based strategy (short the "loser" companies). In addition, the existence of activist short sellers partially balances a market dominated by long only funds and serves to "weed out" the figurative "bad apples". Compared with the SEC (U.S. Securities and Exchange Commission), adopting the data of the activist short sellers has many benefits:

- Activist short sellers tend to be more proactive in their strategy of "early detection," aiming to uncover financial weaknesses before they become apparent. A notable example is the Evergrande bond default in September 2021. Long before the crisis, Citron Research had already exposed issues of accounting fraud and bribery at Evergrande as early as 2012, showcasing the foresight and financial expertise of activist short sellers, who seek to identify and act on these risks well ahead of the broader market (Agyei-Boapeah (2019), Almeida (2022)).
- The activist short sellers in this study are characterized by their professionalism, extensive experience, and expertise in industry-specific knowledge and accounting/auditing. A key example is their role in uncovering the scandal at Wirecard AG, a leading German fintech firm until its collapse in June 2020. Short sellers, along with journalists, raised concerns about Wirecard's accounting practices as early as 2008, and again in 2015, 2016, and 2018, consistently highlighting doubts about the company's financial integrity long before its downfall (Jo et al. (2021)). Notably, Leo Parry of Ennismore Fund Management initiated a short position against Wirecard in 2008, due to concerns about its financial disclosures. His skepticism was later reinforced by Dan Yu of Gotham City Research, who raised further alarms regarding legal issues within Wirecard's payment processing operations (McCrum (2020)).
- Activist short sellers demonstrate a strong "skin in the game" mentality, driven by the prospect of substantial financial gains from their short selling activities, often amounting to millions or even billions of USD. They also face significant costs and the potential for substantial losses if their analysis proves incorrect, as short selling inherently carries high risk. A wrong call can result in unlimited financial losses if the stock price rises, making their involvement in short selling both highly profitable and risky. These motivations far exceed traditional compensation structures like regular government salaries. Their potential rewards are directly tied to the success of their short positions, giving them a higher stake and stronger incentive to uncover and act on financial weaknesses in targeted companies.

In our research, we define "fraudulent companies" as those which have experienced a significant stock price decline after being targeted by activist short sellers, or have faced SEC penalties due to internal fraudulent activities. Our study focuses on U.S. listed companies selected by prominent activist short sellers, examining their financial statements to identify patterns in the financial characteristics of these short-selling targets. The goal is to shed light on the "black box" strategies used by activist short sellers to detect potential corporate fraud.

Short-selling targets and their annual financial reports are publicly available, and we can utilize this data in machine learning – especially supervised learning – for designing algorithms to mimic the professional financial analysis experts. Our aim is to provide valuable insights for investors, analysts, and students seeking to extract deeper knowledge and uncover systematic patterns in financial data. By highlighting how machine learning can assist in financial analysis, we hope to show how these tools can enhance human performance and decision-making in the field of finance, enabling more informed and precise analyses of complex financial trends.

Sophisticated activist short sellers perform extensive research to validate their hypotheses,[9] combining this with sharp business acumen, broad financial expertise, and the experience necessary to assess a company's financial health. Our study focuses on structured data, such as annual corporate financial statements. For unstructured data found in footnotes, management outlooks, and other public information that could suggest potential financial fraud, we plan to explore these areas in future research, as they hold valuable insights often overlooked in traditional financial analyses.

The main contributions of this paper are the following: (1) We manually collect the short-selling targets of eight prominent activist short sellers (Muddy Water Research, GMT Research, etc.) to label the dataset for training multiple machine learning algorithms. (2) We investigate whether including financial features from documented asset pricing factors, accounting models, and financial variables/ratios in the training dataset contributes significantly to the performance of our models. These models include Logistic Regression, K-Nearest Neighbors, Decision Trees, Random Forest, Support Vector Machines, Artificial Neural Networks, Boosting methods (AdaBoost and XGBoost), and LightGBM Regression. (3) We propose the "Polytope Fraud Theory" (PFT), which enhances the understanding of common financial and accounting fraud issues. (4) We propose the "Unified Investor-Protection Framework" (UIPF), which offers a comprehensive approach by categorizing fraud prevention strategies at the macro, *meso*, and micro levels. Additionally, the framework introduces a practical Investor Protection Checklist designed to help investors avoid stocks involved in financial fraud, offering a clear and structured method for assessing risks in publicly listed companies.

The remaining parts of this paper are organized as follows. Section 2 presents a review of financial fraud detection, short selling, and machine learning in the accounting and finance disciplines. Section 3 outlines our data collection, preparation, and cleaning methods. Section 4 details our methodology, while Section 5 presents our empirical findings. Section 6 introduces the Polytope Fraud Theory and the Unified Investor-Protection Framework, which together provide a structured approach

[8] One of famous A.I. failures is Zillow Group Inc., an American online real-estate marketplace company that used A.I. technology to predict property prices. For more information, see https://www.wsj.com/articles/zillow-offers-real-estate-algorithm-homes-ibuyer-11637159261.

[9] Activist short sellers conduct due diligence with suspected fraudulent companies, investigating their customers, suppliers, and competitors, and collecting information from the tax authority.

to identifying and mitigating fraud in financial markets. Section 7 presents our conclusion.

2. Literature review

2.1. Motivations for fraud

The essence and contributing factors of fraud have been widely examined in the academic literature. Marks (2012) explores the behavioral elements involved in committing fraud and introduces the Fraud Pentagon, adding "arrogance" as a key factor. Arrogance is defined as an attitude of superiority, where individuals believe internal controls do not apply to them. Jo et al. (2021) studied the collapse of Wirecard AG, identifying factors such as pressure on management to meet expectations and the consensus to maintain a facade despite underperforming business growth. Similarly, Magdalena and Dananjaya (2021) analyzed the influence of CEO capabilities and arrogance on fraudulent financial statements. Using linear regression, they discovered that CEO traits—such as age, tenure, and political connections—play a significant role in fraud.

Sihombing and Cahyadi (2021) researched the influence of Fraud Diamond elements in detecting financial statement fraud in companies listed in the S&P Asia-Pacific region from 2017 to 2019, employing empirical methods to test contributing factors. Mvunabandi (2022) conducted a theoretical review to extend the debate on fraud theories, proposing a robust fraud combination theory aimed at aiding forensic and external auditors in fraud detection. Dewi and Pinasthika (2023) examined the impact of the Fraud Diamond – specifically pressure, opportunity, capability, and rationalization – on financial statement fraud in insurance companies, using 55 observations from 2016 to 2020 and statistical analysis via STATA, showing variation in fraud occurrence across sectors.

Yarana (2023) provided empirical evidence on financial statement fraud occurring in Thailand, aligning with Fraud Diamond Theory. The study found that external pressures, such as financial targets (ROA), rationalizations like accrual (ACCRUAL), and industry risk (IND), significantly influence financial statement fraud on the Stock Exchange of Thailand. The analysis employed descriptive statistics and logistic regression.

Cho et al. (2024) explored the relationship between the content of earnings information and the use of "trust words" in the MD&A section of 10-K reports. Their findings suggest that the use of trust words is associated with negative outcomes, implying that trust words serve as an inverse measure of actual trust. Lastly, Ni et al. (2024) examined the design of a robust credit card fraud detection system. Their proposed model, which leverages historical transaction data, demonstrates the effectiveness of CGEA, MSEFBoost, and SOBT in improving fraud detection capabilities.

Motivations for financial statement fraud generally fall into two categories: contracting incentives, such as management bonus programs tied to accounting outcomes, and capital market incentives, where companies seek favorable financing terms or artificially inflate asset values to benefit from insider trading (Young, 2020). In rarer instances, companies may intentionally deflate their market value to facilitate favorable takeover conditions. Each motivation reflects different pressures faced by management to meet targets or manipulate perceptions in the capital markets.

The presence of accounting-based compensation (e.g., EBIT, net income, revenue growth) tends to result in more accounting irregularities than the absence of such compensation. Healy (1985) showed that the choice of accounting method and overstating or understating financial statements are determined by the management in a large part, which is likely to have agency problem issues. Efendi et al. (2007) documented that the likelihood of "restatement" required by the SEC increases significantly when Chief Executive Officers (CEOs) have sizable holdings of in-the-money options. This implies that the direct link between senior executives' personal interests and stock price performance might lead to potential accounting malpractices to boost, or at least sustain, the stock price. Peng and Röell (2008) also indicated that the existence of executive options increases the likelihood of securities-related litigation. Call et al. (2016) reported similar findings.

The pursuit of desirable equity prices or more favorable financing terms creates another strong motivation for financial statement manipulation. Companies may seek to present a stronger financial position to attract investors, boost stock prices, or secure better borrowing conditions. Dechow et al. (1996) revealed that a company might inflate its earnings to obtain a better price for new equity issuances or to lock in a lower interest rate. Burns & Kedia, 2006 found that firms are willing to adopt aggressive accounting practices to reduce the costs of financial distress. Perry and Williams (1994) argued that, if managers want to acquire the company/division through Management Buyout (MBO), they tend to understate the corporate earnings. Beneish (1997, 1999a) posited that net insider selling will appear when earnings are inflated. This suggests that managers might overstate their financial statements before they sell their shares. Peng and Röell (2008) uncovered similar results, showing that securities lawsuits increase when insiders are not selling. This implies that insiders anticipate that they might be charged with financial misconduct, so they sell more shares or options during class-action litigation periods.

2.2. Fraud triangle theory and Fraud Diamond theory

2.2.1. Summary of the two theories

There are various leading factors or conditions that most fraud events possess. The two most cited related theories are Cressey's (1953) "Fraud Triangle Theory" (FTT), and Wolfe and Hermanson's (2004) "Fraud Diamond Theory" (FDT). Donald Cressey, a criminologist and former student of Edwin Sutherland (Sullivan coined the term "White-collar crime") proposed the foundation theory of fraud[10] based on (a) Pressures, (b) Opportunities, and (c) Rationalizations or Attitudes.[11] The major difference between FTT and FDT is that FDT adds a fourth element to the FTT, which is the "Capability" to commit fraud.[12]

$$\text{FTT} : \text{Fraud} = \text{f}\,(\text{Pressure}, \text{Opportunity}, \text{Rationalization}).$$

$$\text{FDT} : \text{Fraud} = \text{f}\,(\text{Pressure}, \text{Opportunity}, \text{Rationalization}, \text{Capability}).$$

Research shows that management is under pressure when it has to meet analysts' forecasts, confront poor performance, or source external financing. Loebbecke et al. (1989) and Bell et al. (1991) established that, if a company is experiencing growth that is below the industry average, management may manipulate the earnings to improve corporate output. Rosner (2003) pointed out that failing firms are more likely to inflate the income statement to hide financial distress based on accruals (decreased cash flows or more accounts receivables). Davidson (2016) concluded that managers commit balance sheet-related fraud when the market-wide default risk is high, and their firms are in greater financial distress. In addition, managers commit income statement-related fraud when their firms' stock price is sensitive to their idiosyncratic earning performance.

[10] Cressey never referred to the three elements of fraud as the "fraud triangle". It is Albrecht (1991) who completed Cressey's trust violation theory and named it "the fraud triangle", as he considered the three elements of fraud similar to the "fire triangle".

[11] The three elements of fraud triangle theory can be explained as (1) Incentive: One wants to, has to, or needs to commit fraud; (2) Opportunity: The system has a weakness, so fraud is possible; (3) Rationalization: There exists an assessment that committing fraudulent behavior worth the risks.

[12] The added fourth element in the fraud diamond theory, "Capability", can be considered as "One is the right person with the necessary skills and abilities to commit fraud".

Opportunities for fraud arise in conditions of weak internal control (for example, poor corporate governance and lack of supervision), lack of external monitoring (auditors being undercompensated and overworked), and inadequate accounting policies (poor separation of duties and poor documentation of processes). Dechow et al. (1996) indicated that companies with fewer independent boards, more unitary structures for the chairman and CEO, and fewer outside block holders are more likely to manipulate earnings. Farber (2005) affirmed that fraudulent firms tend to have fewer independent board members, fewer audit committee meetings, fewer financial experts on the audit committee, and a higher percentage of CEOs as the Chairman than normal firms. Researchers also showed that auditor tenure is related to earnings quality (Iyer & Rama, 2004; Myers et al., 2003). Carcello and Nagy (2004) showed that auditor industry specialization and financial irregularities are negatively correlated with each other.

Rationalization indicates that the firm's management must accept immoral ideas before engaging in unethical behavior. Gillett and Uddin (2005) showed that the CFO's attitude toward fraudulent behavior influences accounting malpractices. Howe, and C., Malgwi. (2006) argued that the gap between pressure and opportunity is filled when individuals rationalize behaviors, for example, employees' lack of personal integrity or moral reasoning (McCulloch and Pitts, 1988; Rae & Subramaniam, 2008).

Capability – the fourth element in the "fraud diamond" – describes the necessary skills or abilities to commit fraud. Albrecht et al. (1995) found that capability is essential, especially for large-scale and long-term fraud. Albrecht et al. (1995) also argued that only a competent person who thoroughly understands the internal control and auditing process could implement the fraud. Wolfe and Hermanson (2004) similarly observed that position, intelligence, ego, and stress are the supporting elements of capability: Individuals within an organization who are intelligent enough to understand where the weaknesses are can exploit them.

Abundance of opportunity also plays a large part in the psychology that motivates certain fraudulent activities, especially for the "high capability" fraud.[13] Even when external deterrents are lacking, strong internal controls can prevent individuals with high capability from exploiting weaknesses (Murphy & Dacin, 2011). Knowing that they might get caught and punished deters the "high capability" managers from committing fraud (Carland et al., 2001; Hirschey et al., 2009; Kassem & Higson, 2012; Rose et al., 2015; Baker et al., 2020).

Albrecht and Albrecht (2004) categorized six signs of fraud: (1) accounting anomalies, (2) internal control weakness, (3) analytical anomalies, (4) extravagant lifestyles, (5) unusual behaviors, and (6) tips & complaints. Although these symptoms are observed frequently in fraudulent companies, it is still complicated to formulate a specific audit plan to uncover fraud.

2.2.2. Illustration with the case study of Enron

Based on the book *Man-made catastrophes and risk information concealment: 25 case studies of major disasters and human fallibility* (Chernov & Sornette, 2016), we summarize the case of Enron to illustrate the Fraud Triangle Theory and the Fraud Diamond Theory.

Founded in 1985, Enron was one of the largest companies in the United States before filing for bankruptcy, on December 2, 2001 (Chapter 11). Enron focused on wholesale merchants, commodity markets, gas transmission systems, and retail energy services, and had around 3500 domestic and foreign subsidiaries and affiliates. The annual revenue soared from 9 billion to 100 billion dollars between 1995 and 2000. After the Enron accounting fraud was uncovered, Enron's stock price plummeted from 90 dollars per share to less than 1 dollar, evaporating $11 billions of (59,000) shareholders' wealth. Several pension funds experienced 2-billion-dollar losses, and 20,000 creditors received just 14 to 25 cents on every dollar they lent to Enron. The Enron scandal, being the largest firm bankruptcy in the American history, led to the dissolution of Arthur Andersen, one of the five largest auditing firms in the world at that time.

According to the Fraud Diamond Theory, financial fraud can be explained through four key elements that are found to present in the case of Enron:

Incentive: After two debt-financed merger acquisitions, Enron had an enormous debt of 4.3 billion dollars. To top it off, the deregulation of the energy sector in the late 1980s meant Enron lost its exclusive rights to the gas transportation pipeline in the natural gas market. These two revenue-impacting issues created a huge pressure for Enron's management team, and the company reported a 79-million-dollar loss in the late 1980s. After Jeffrey Skilling became the CEO, Enron's compensation structure was radically changed. The employees' pension fund was heavily invested in Enron's stock and Enron's aggressive payment structure motivated employees to increase Enron's capitalization – by any means. Senior management created Special Purpose Entities (SPEs) in their own names, which could be used to hide the corporate debt and to transfer the money into their own pockets. As this enriched them as individuals through self-dealing, it presented a material conflict of interest with company shareholders.

Opportunity: The federal deregulation of the energy sector during George H. W. Bush's U.S. presidential term created opportunities to expand Enron's commodity trading business. Due to a strong tie[14] with the government, Enron successfully lobbied to remove regulations on over-the-counter energy derivatives. In addition, the SEC's approval of the mark-to-market (MTM) accounting method for energy companies in 1992 allowed Enron to report expected benefits from future transactions into current income, inflating its revenue from $6.3 billion to $100.8 billion by 2000. Arthur Andersen, Enron's auditing firm,[15] assisted Enron to implement aggressive accounting to increase revenue. Furthermore, Enron had no internal auditing department, and many of the financial managers of Enron were former executives of Arthur Andersen. The board directors at Enron were compensated with salaries significantly higher (twice as much as the top range of what directors at a typical public company earn) for their roles, despite having a limited grasp of the company's operations. Corrupt politicians, greedy auditors, and "silent" board directors provided weak control and created ample opportunities for fraudulent activities.

Rationalization: Under the leadership of Jeffrey Skilling, Enron's corporate culture changed dramatically. Analysts argue that it became less of an energy company and more like an investment bank. Enron became known for rewarding and promoting employees who secured large new deals, while applying considerable pressure on those who did not achieve similar successes. The remuneration system was redesigned to focus on making big deals, regardless of quality. The company hired aggressive and smart young graduates and gave them generous salaries,[16] and reportedly fired those in the bottom 15 % quantile. An

[13] Intuitively speaking, if the opportunity is preferable (i.e., it is easy to implement the fraud), less capability is needed to commit the fraud. Similarly, if the opportunity is weak, the manager needs a strong capability to commit accounting fraud.

[14] Enron's chairman Kenneth Lay was the co-chairman of Bush's 1992 re-election committee, and he had made large monetary contributions to the Bush presidential campaign. For more information, see https://www.economist.com/unknown/2002/01/11/bush-and-enrons-collapse.

[15] Arthur Andersen earned one million dollars per week for auditing, and its consultation services accounted for 70 % of the total payment from Enron.

[16] The base salary at Enron was 51 % higher than its peer group. In addition, employees' bonus payments were 383 % higher, and stock options were 484 % higher (Chernov and Sornette 2015).

auditor who dared to question Enron's financial reports was fired.[17] No one in the investment banks marked Enron as a "sell" target because many banks had businesses with Enron. The culture of "success at any cost" and "report only good news" blinded internal employees and external financial institutions, distorting their professional ethics (Chernov & Sornette, 2016).

Capability: Kenneth Lay (chairman of Enron, Ph.D. in economics), Jeffrey Skilling (CEO of Enron, Harvard University MBA), and Andrew Fastow (CFO of Enron, Northwestern University MBA) were all highly educated and intelligent. Andrew Fastow engineered a network of 3000 out-of-balance special purpose entities (SPEs) to hide the enormous debt and transfer huge losses resulting from merchant investments to third-party companies. To accomplish favorable financial statement results, they unconsolidated the unprofitable international subsidiaries and affiliates. They committed money laundering, filed fraudulent income tax returns, and transferred corporate assets to their own pockets through complex and opaque accounting structures. The Enron scandal was only uncovered after many years of fraudulent activities, through one of their employees, Sherron Watkins, who became a whistle-blower in 2001. Before that, Enron was an award-winning company often cited as case study material for innovation in business schools across America.

The Enron scandal has led to a huge crisis of belief in the trustworthiness of the whole U.S. capitalist system, since it casted doubt on the accounting practices and financial reports of U.S. listed companies. Thus, the American Institute of Certified Public Accountants (AICPA) issued the Statement of Auditing Standards No.99 (SAS No. 99), and U.S. congress passed the Sarbanes-Oxley Act of 2002 (also known as the "Public Company Accounting Reform and Investor Protection Act"), in response to the collapse of the Enron empire (Skousen et al., 2009). The Sarbanes-Oxley Act aims to increase financial information accuracy, strengthen corporate governance, and increase the severity of financial fraud penalties for corporate management (Chen et al., 2015). The Act also created a new agency, the Public Company Accounting Oversight Board (PCAOB), to regulate and oversee accounting firms' independence and internal controls.

2.3. Market efficiency versus inefficiency

Market pricing efficiency is a basic tenet of financial economics (Fama, 1970). In this spirit, Grossman and Miller (1988) stated that the stock price would move toward fundamental value due to arbitrage forces. Lee et al. (1999) also indicated that the stock price and fundamental value form a co-integrated system, and arbitrage will force them to converge. However, with the existence of various mechanisms (bounded rationality, limits of arbitrage, noise, etc.), the price can deviate significantly from fundamental value for quite a long time. This is known as "mispricing". Dechow and Sloan (1997), Espahbodi et al. (2001), and Ang and Ma (2001), determined that analysts' earnings forecasts are generally over-optimistic. Black (1986) stated that noise trading contributes to the long-time divergence of price and intrinsic value, and unreasonable price movements. Hüsler et al. (2013) revealed from laboratory experiments that the over-optimistic expectations on price would result in positive feedback favoring bubble formation.

Behavioral finance theory points to the limits of arbitrage and the presence of irrational investors as the reasoning that the market is inefficient. Shleifer and Vishny (1997a, 1997b) proposed that arbitrage costs deter the price from approaching its intrinsic value. Specifically, there are three kinds of arbitrage costs: (1) trading costs: brokerage fees, order implementation cost, or other fees that are related to building or closing trading positions; (2) holding costs: the costs of sustaining the duration of the position, for example, short selling interest; and (3) information costs: costs associated with accessing, analyzing, and monitoring information. In addition, Edwards (1968) suggested that investors tend to undervalue new information when they update their information set. Barberis et al. (1998) indicated that the "conservative bias", which means prices will slowly adjust to new information, might result in an underreaction by investors and generate a "momentum effect". Daniel and Hirshleifer (2015) asserted that overconfident investors, including experts and professionals, contribute to excessive trading and disagreement between price and value. Various psychological biases lead to mispricing.

Sornette (1999) proposed that heterogeneous investors with repetitive interactions have the tendency to imitate others, resulting in "herding" and crowd effects. Cooperative herding and social imitation can emerge out of equilibrium properties. Sornette (2004) noted that self-reinforcing imitation among noise traders is one among many mechanisms in financial economics leading to nonlinear positive feedback, which leads to unsustainable super-exponential price dynamics – a hallmark of financial bubbles. For instance, these transient super-exponential price dynamics can be used with the LPPLS (log-periodic power law singularity) model to diagnose developing bubbles and forecast their end times (Sornette, 2004; Sornette & Cauwels, 2015; Ardila-Alvarez et al., 2021; Zhao & Sornette, 2021).

2.4. Agency problem and investor protection

Agency theory, one of the oldest theories in management and economics literature, discusses problems due to the separation between the owners and managers within a firm (Daily et al., 2003; Wasserman, 2006). Adam Smith discussed in his book *The Wealth of Nations* that, if an organization is managed by a person other than the real owner, then the people might not work for the owner's benefit. Berle and Means (1932) observed that many large U.S. firms had dispersed ownership, leading to the separation of ownership from control. They argued that self-interested agents might use the firm's property for their benefits, creating conflicts between the principals and agents.

In the modern era, the agency problem is not limited to principals and agents – it extends to other parties such as creditors, major shareholders, and minor shareholders (Panda & Leepsa, 2017). Researchers have categorized three agency problems: (1) an agency problem arises between the principal and agents due to information asymmetry and divergent risk-taking attitudes (Jensen & Meckling, 1976; Ross, 1973); (2) an agency problem exists between the major and minor shareholders (Gilson & Gordon, 2003; Shleifer & Vishny, 1997a, 1997b)[18]; (3) an agency problem occurs between owners and creditors: higher-risk investment decisions by owners can be unsupported by existing creditors (Damodaran, 1997).

Jensen and Meckling (1976) argued that a misalignment of interests between the agent and the principal can lead to agency conflicts and related costs. There are three agency costs: (1) monitoring costs, which are the costs associated with monitoring and assessing the agent's performance; (2) bonding costs, which are the costs incurred to set up and operate according to predetermined contractual obligations, and (3) residual costs, which are losses due to inefficient managerial decisions. Williamson (1988) claimed that residual cost is the key component of agency cost.

The lack of effective legal restrictions on the ability of corporate insiders (managers or controlling shareholders) to divert corporate wealth to themselves, known as "self-dealing", is a major investor protection problem (Grossman & Hart, 1983). Abundant research indicates

[17] Carl Bass, one of the auditors from Arthur Andersen, questioned the mark-to-market accounting method, but he was fired, and his CPA membership was canceled by the Texas State Board of Public Accountancy (TSBPA), which was under the leadership of Mike Conaway, mutual friend of both George Bush and Kenneth Lay.

[18] Shleifer and Vishny (1997a, 1997b) indicated that an agency problem can also occur between outside investors and controlling shareholders, who have almost full control of corporate management.

that differences in legal investor protection across countries (which restrict the ability of insiders to gain "private benefit of control") determine investor confidence in markets, and consequently the differences in their development (La Porta et al., 1997, 1998; Shleifer & Vishny, 1997a, 1997b; Shleifer & Wolfenzon, 2002). Djankov et al. (2005) documented the differences in legislation between countries in protecting minority shareholders and creditors. They indicated that the ability of different legal bases to restrict the self-dealing problem impacts the development of the corresponding financial markets. Shleifer and Wolfenzon (2002) also revealed that investor protection shapes external finance. In addition, Shleifer et al. (2008) proposed the "Legal Origin Theory", suggesting that common law countries and regions (U.K., U.S.A, Singapore, Hong Kong, etc.) protect outside shareholders and outside creditors better than civil law countries and regions (France, Germany, Italy, Japan, People's Republic of China, etc.), and thus financial markets in common law countries become larger and better functioning, with more sources of external financing.

2.5. Short sellers

Research indicates that short sellers are sophisticated investors. Typical short sellers include professional traders, hedge fund managers, portfolio managers, etc., and they play essential roles in the dynamics of price discovery, stock market efficiency, and corporate manager discipline (Boehmer et al., 2008; Christophe et al., 2004; Diether et al., 2009). Short sellers, based on the underlying economics or fundamental analysis of stocks, decide whether a stock is overvalued or not, and act accordingly. Studies suggest that short selling constitutes around 20–31 % of trades in the U.S. market (Chen et al., 2019). The short seller's business model is based on the expectation of future drops in the target firms' stock price. Short sellers profit from the strategy of "selling high and buying low", which means they buy back the securities at a lower price and return them to the broker they originally borrowed from at a higher price.

There are two types of short sellers: passive short sellers and activist short sellers. The basic difference is that activist short sellers declare their short position to the public, whilst passive short sellers keep quiet and tend to implement market-neutral arbitrage strategies. Portfolio managers may execute short selling of stocks to hedge against any downside risk of a long position (Ederington, 1979), or short the stocks of the acquiring firm while building the position of the target firm in merger and acquisition events (Baker & Savaşoglu, 2002). Passive short sellers usually build positions without announcing their actions, even after they close their positions. In contrast, activist short sellers publicize their opinions about target companies through detailed public reports online, accusing the target firms of financial statement fraud (e.g., accounting irregularities, earning manipulations, fake transactions, etc.). Activist short sellers aim to push the stock price down, as their short positions have been built in advance. Aiming on this, they not only acquire large short positions,[19] but also persuade other long-side buyers (mainly institutions) to cut their long positions.[20]

In general, although some activist short sellers may take controversial positions in applications, they publish reports with sufficient evidence and proceed with information more efficiently than passive short sellers, since activist short sellers often conduct extensive research, while passive short sellers often short stocks based on simple factors. In addition, activist short sellers tend to create stock price decrease actively (i.e. they might trigger stock price crashes of some companies) rather than for tax or hedging purposes (Brent et al., 1990). Accordingly, activist short sellers, who are unconstrained by the supply in the equity-loan market, take more risks than passive short sellers.

There are three significant risks that activist short sellers face:

- First, whether the stock price will decline enough to at least compensate for the additional costs (e.g., borrowing costs, transaction costs, and information costs) and corresponding risks (e.g., market risk). In addition, short sellers have to face the possibility of a "short squeeze", whereby the stock price increases instead of decreasing, so that they have to buy back the stock at a higher price than their selling price in order to return the stock to the broker. In this case, they not only have to bear the costs, but also suffer from losses associated with their position.[21] In theory, a short sellers' maximum loss is unlimited since the stock price can increase to unlimited levels (Diamond & Verrecchia, 1987).
- Second, whether the shorted company will sue the activist short sellers for misleading in a material respect (i.e., market manipulation). If the activist short sellers engage in market manipulation (intentionally misleading the public), they may face criminal prosecution. For instance, in the British jurisdiction, under Sections 89 and 92(1)(b) of the Financial Services Act (FSA), anyone who has engaged in market manipulation could be sentenced to a maximum of seven years' imprisonment and incur an unlimited fine (Durston, 2021).
- Third, when short sellers "touch other people's cake", they may, of course, invite trouble into their personal lives. That is also why many activist short sellers refuse to reveal their identity – to protect themselves and their connections. Most of the sophisticated short sellers which we reviewed have revealed their names and contact details. Consequently, some of them have received hundreds of emails threatening their lives or their safety, or that of their family members.[22]

Research indicates that short sellers have abilities superior to auditors and financial analysts in detecting financial statement fraud by exploiting public information (Ackert & Athanassakos, 2005; Aitken et al., 1998). Their existence in the financial market accelerates the discovery of overvalued stocks, increasing market efficiency, and restraining corporate managers from potential fraud activities (Boehmer et al., 2008; Christophe et al., 2004, 2010; Henry et al., 2015). They also play an essential role as a counterbalance to excessive market optimism (Keshk & Wang, 2018; McNichols & O'Brien, 1997). Research also indicates that short selling restrictions and constraints slow down the incorporation of negative private information and thus reduce market efficiency and hedging effectiveness (Brunnermeier & Oehmke, 2014; Choy & Zhang, 2019; Danielsen & Sorescu, 2001; Hasan et al., 2015).

Short sellers tend to target firms with a poor fundamental ratio, large market value, and high institutional shareholding (Dechow et al., 2001). In addition, short sellers are more activist in high-growth and financially opaque firms, which are susceptible to overvaluation (Grullon et al., 2015). Dechow et al. (2001) found that short sellers heavily target companies with significant deterioration in their fundamentals. Kecskés et al. (2013) also demonstrated that stocks with lower credit ratings are more likely to have ratings downgraded or develop distressed situations – such stocks constitute desirable targets for short sellers. Desai et al.

[19] Activist short sellers might borrow a lot of stocks and suddenly dump them into the market to cause panic among investors.

[20] Activist short sellers disclose their detailed short-selling theory with conclusive facts and strict logic to persuade other investors to terminate their long positions.

[21] For example, short sellers had shorted GameStop (GME) stock since 2020, while the GameStop stock price increased more than 1700 % from January 1, 2021, to February 2, 2021.

[22] For instance, after Citron released the short-selling report of GameStop, Andrew Left received anonymous threatening phone calls using fake voices and had strangers show up at his personal residence. Citron's twitter account was hacked. Thus, Andrew Left decided to stop issuing short-selling reports and his fund has turned to implementing long-only investment strategies since January 2021.

(2006) observed that short sellers tend to avoid small firms, a behavior that might be due to high liquidity risk and the difficulty in borrowing their stocks.

Current regulations in the U.S. relevant to activist short sellers mainly come from the Security Exchange Act of 1934. According to Section 78i, any “false or misleading statement” to facilitate stock transactions is considered “manipulation of security prices.” The Act grants the SEC the power to investigate and issue severe penalties to punish anyone suspected of manipulating stock prices through a “false or misleading statement”. However, the burden of proof lies with the SEC to demonstrate reasonable grounds that the short sellers were using false and misleading information to gain special advantages over transactions.

2.6. Financial statement fraud detection

Machine learning models are effective and accurate in classifying performance (Feroz et al., 2000; Kotsiantis et al., 2006; Wu et al., 2008; Perols, 2011). Their decision-making process is more objective, and more data can be efficiently processed. Unlike traditional statistical methods that require time-consuming and expensive manual detection, machine learning algorithms such as logistic regression, artificial neural networks (ANN), fuzzy logic, and ensemble-based methods have more computing power and can find subtle non-linear relationships among data (Green & Choi, 1997; Kirkos et al., 2017; Lokanan & Sharma, 2018; Perols, 2011). Yue et al. (2007) reviewed the existing literature on machine learning applied to fraud detection before 2006 and claimed that the most researched methods of fraud detection are classification-based, which Sharma and Panigrahi, 2013 and West and Bhattacharya (2016) confirmed in more recent literature reviews.

A common thread in previous research is to find indicators related to potential fraud, known as “Red Flags” (Cecchini et al., 2010). However, Green and Choi (1997) indicated that there are no formal theoretical indicators of fraud – people usually choose some financial attributes using expert judgment, but on an ad-hoc basis. Green and Choi (1997) used 46 fraudulent and 49 non-fraudulent companies to train their ANNs to identify financial statement fraud, with five revenue-related ratios from previous audit assessments research. They obtained a 72 % accuracy. The major limitation of their study is that the neural network seems to be a “black box” approach providing little insight, and the researchers ignored the unbalanced nature between fraud and non-fraud companies within the sample. Summers and Sweeney (1998) used 51 fraudulent and 51 non-fraud companies to train the cascaded logit model with financial variables and obtained the same 72 % accuracy. The main weakness of their work was the lack of out-of-sample validation. Kotsiantis et al. (2006) used Decision Trees, ANN (Artificial Neural Network), SVM (Support Vector Machines), and nearest-neighbor methods to classify 164 Greek firms (41 fraud and 123 non-fraud) with financial statement data and variables and achieved above 90 % accuracy. However, they did not conduct out-of-sample tests.

Hoogs et al. (2007) introduced a genetic algorithm to classify 51 fraud companies (using SEC accusation data) from 339 peers with similar industry and size (revenue) using 76 comparative financial metrics and ratios. Their genetic algorithm correctly classified 44 % of the allegedly fraudulent companies (True Positive). Dechow et al. (2009) implemented a logistic regression model to study 293 fraud and 79,385 non-fraud cases, and their model was able to classify 70 % of the fraud firms (True Positive) and 84.9 % of the non-fraudulent firms (False Negative) in the testing sample. Cecchini et al. (2010) used an SVM to classify the 137 fraudulent company years and 3187 non-fraudulent firm years using SEC restatement data from 1991 to 2000 and obtained 80 % of fraud cases (True Positive) and 90.6 % of non-fraud cases (False Negative) in the test set.

Sharma and Panigrahi, 2013 examined and compared data mining applications and fraud detection systems in the literature and found that Logistic Models, ANN, Bayesian Networks, and Decision Trees’ data mining techniques are most widely applied to provide solutions to the problems of identification and classification of fraudulent data. West and Bhattacharya (2016) categorized the research from 2004 to 2015 and discovered that supervised learning tools are more frequently used than unsupervised learning tools. Albashrawi (2016) reviewed 41 financial statement fraud articles and pointed out that Logistic Regression, Decision Trees, SVM, Neural Networks, and Bayesian Networks have been widely used (in more than 50 % of the studies) to detect financial fraud.

Additionally, in the field of fraud and anomaly detection, various methods such as case studies, lab experiments, and computational models have been employed. For instance, Wei et al. (2023) explored the detection of malicious chargebacks in online games, discovering that while deep learning models were less efficient compared to traditional ones, they proved effective in reducing losses for online game companies. This highlights the evolving role of AI and machine learning in addressing complex fraud detection challenges across different industries, by applying methods such as deep learning models, decision trees, and K-Nearest Neighbors. Alarfaj et al. (2022) research credit card fraud detection, focusing on public data accessibility, class imbalance, changes in fraud nature, and high false alarm rates. They show that a robust classifier can handle fraud’s changing nature, emphasizing the importance of input data type in driving different machine learning methods, including extreme learning methods, decision trees, and K-Nearest Neighbors. Hsin et al. (2022) focus on detecting fund transfer fraud using features like recency, frequency, monetary value, anomalies, behavior, and segmentation. Using raw transaction data from a bank and fraudulent account records from the National Police Agency, they find that time-inhomogeneous properties of features cause instability in popular boosting classifiers like XGBoost and LGBM. Ali et al. (2023) develop a financial statement fraud detection model utilizing data from publicly available financial statements of firms in the MENA region. They use the ensemble technique and the XGBoost algorithm to develop a prediction model, investigating various ML techniques in detecting financial statement fraud. Dyck et al. (2021, 2024) investigate whether the detected instances of corporate fraud represent the entirety of the problem or merely the “tip of the iceberg.” They leverage the natural experiment created by the collapse of Arthur Andersen, one of the Big Five accounting firms, to estimate the proportion of corporate fraud that remains undetected. Arel et al. (2023) study the influence of fraud diamond capability measures on the probability of committing fraud and test the effectiveness of the fraud diamond model in identifying fraud. Their experimental task involving 91 undergraduate business students shows that capability components like ego, intelligence, and stress coping significantly influence the likelihood of committing fraud. Achakzai and Peng (2023) detect corporate fraud in China using Machine Learning models. Utilizing data from CSMAR (China Stock Market and Accounting Research), they develop an advanced fraud detection model that dynamically combines multiple individual classifiers to improve accuracy, introducing the Dynamic Ensemble Selection algorithm. This approach utilizes methods such as logistic regression, SVM, and Random Forests. to achieve more reliable final predictions by selecting the most appropriate model for each case based on the characteristics of the data. This method enhances detection accuracy by leveraging the strengths of different algorithms in a dynamic and adaptive manner. Chen and Zhai (2023) analyze the characteristics of financial fraud data scenarios, using listed companies in the A-share market from 2012 to 2022 as samples. They consider six dimensions: asset quality, profitability, solvency, operating ability, cash flow analysis, and market environment. They provide mechanistic evidence supporting the applicability of the bagging ensemble strategy in various scenarios, thereby enriching the research on ensemble learning in economics and management. The study leverages advanced machine learning methods, including Random Forest, gradient boosting classifier (GBC), AdaBoost, XGBoost, and LightGBM, to enhance predictive accuracy. Zhang and Dai (2024) investigate accurate and interpretable methods for detecting electricity theft patterns among end-users. Their study demonstrates that the proposed detection models significantly help power operators in conducting audits, minimizing economic losses, and stabilizing power grid operations. They employ advanced machine learning techniques, such as XGBoost, MCNN, and the Attention model.

There are two common problems in most of the previous machine learning research that focused on fraudulent financial statements: missing values and sample imbalance (Rubin, 1976; Van Buuren, 2018). First, deletion is a general method to handle the missing-data samples. That is, if the financial features of a target firm are missing, researchers tend to remove incomplete sample firms, which might lead to biased or incomplete results. Deletion is also the primary reason for the small number of samples used in research. Second, due to the infrequent nature of fraudulent cases, the fraud to non-fraud ratio is actually very small, and there exists a highly imbalanced problem in the real world. However, much of the previous research used a fraudulent to non-fraudulent ratio close to (1:1) (Ye et al., 2019). Therefore, small samples and unrealistic fraud ratios are the two "Achilles' heels" of otherwise excellent machine learning results.

There is an alternative way to detect financial statement frauds, namely, using statistical properties of the reported numbers in financial statements (Sornette, 2004). Frank Benford (1938) noticed a pattern in Newcomb's (1881) dataset that the digits 1 to 9 are not equally distributed in "natural" sets of numbers, for instance in an encyclopedia, and the lower digit numbers appear more frequently than higher digit numbers. The distribution of numbers follows the so-called Benford's law, which derives from the log-periodicity inherent in the number theory; for instance, the base-10 decimal system exhibits a log-periodicity with a preferred scaling ratio of 10 by construction (Sornette, 1998). Researchers collected 20,000 first digits from various sources, including river area, population, constants of physics, newspapers, addresses, death rates, etc. They found that Benford's law is widespread in the natural world. Research also shows that Benford's law appears in the finance and accounting disciplines. Nigrini and Mittermaier (1997) and Guan et al. (2006) showed that Benford's law can also be used in auditing and taxation to detect "cosmetic earnings management" and accounting fraud. Kossovsky (2014) showed that Benford's Law can be applied in forensic fraud detection, tax fraud in international trade, and market collusion.

2.7. Asset pricing factors and accounting "Red Flags"

One way to reduce the overfitting problem is to use domain expert knowledge. The accounting literature indicates that some accounting ratios can be used to detect financial statement fraud, and asset pricing theories show that some factors can be used to construct long/short portfolios. We include these accounting models and quantitative factors in our machine learning features.

Beneish (1999b) showed that sales growth, accruals, leverage, and other factors can be used to detect financial anomalies such as earning manipulation, and he created what is now called the Beneish M-score to classify fraudulent firms. The model covers eight financial features. Summers and Sweeney (1998) also demonstrated that disparities between inventories, return on assets, and other factors can be used to detect financial fraud. Montier (2008) observed that firms manipulating their financial statements might have lower future earnings and expected returns. Therefore, he created the Montier C-score, using six accounting red flags to detect earning manipulations that frequently appear in financial practices. Dechow et al. (2011) developed another model – the fraud score (F-score) model, which is a more recent variant of the Beneish M-score. Their model evaluates fraudulent activities of companies in five areas: accrual quality, financial performance, nonfinancial measures, off-balance-sheet activities, and market-based measures.

To understand how activist short sellers systematically measure the financial position of fraudulent firms, we also included many well-known asset-pricing quality factors that have predictive power for expected returns and reflect corporate financial strength. For instance, Altman (1968) created a *Z*-Score that can be used to detect the bankruptcy risk of a business, which is calculated using financial ratios from financial statements. It is also documented that firms with higher credit risks tend to underperform their market peers in stock price return (Campbell et al., 2008; Ohlson, 1980). Zhang et al. (2010) applied Z-Score in credit rating for listing firms in China.

Ou and Penman (1989) indicated that some financial ratios can lead to expected earnings. Sloan (1996) showed that high-accrual[23] firms are more likely to have downside earnings surprises and that low-accrual firms tend to outperform their competitors. Piotroski (2000) and Piotroski & So (2012) found that the F-score, which combines nine binary financial ratios (covering profitability, efficiency, and leverage in financial statements), can be used to construct portfolios that generate excess return. Back-testing results show that portfolios consisting of higher F-score stocks can persistently generate excess returns after deducting the transaction costs. In addition, portfolios that contain lower F-score stocks tend to underperform the benchmark. Mohanram (2005) created a G-score by comparing eight accounting ratios with the correspondent sector medians. Li and Mohanram (2019) showed that portfolios with higher G-scores tend to have more competitive advantages in relative fundamental strength and can generate excess portfolio returns. Chanos (2003) used "ROIC-WACC" (value-added economic model) to analyze the Enron fraud and set up a short position against Enron (and made a fortune with it). Novy-Marx (2013) showed that firms with higher gross profitability, defined as revenue minus COGS (cost of goods sold) scaled by total assets tend to outperform firms with lower gross profitability. Asness et al. (2019) combined 21 financial features to create the Quality-Minus-Junk (QMJ) factor. Their research shows that high-quality stock portfolios beat low-quality stock portfolios. We added standard accounting red flags (e.g., short-term debt as a percentage of total debt) and financial ratios (e.g., ROA, gross margin, etc.) to the feature inputs and utilized 89 financial features for training the machine learning algorithms.

The above accounting-related factors, ratios, and variables are well-known in the asset-pricing and accounting literature. In addition, many of the famous quantitative long/short hedge funds[24] or asset management companies use the mentioned factors to construct investment strategies, by building long or short positions based on the rankings of the stock according to those factors.[25] In the present research, we include these passive short-selling accounting factors, accounting red flags, and fraud detection models, trying to unveil some of the hidden recipes of the activist short sellers.

3. Data preparation, data cleaning, and features selection

3.1. Data preparation and data cleaning

We outline the steps of our research process in Fig. 1.

To avoid sample bias, we chose eight well-known activist short sellers (Muddy Waters Research, GMT Research, Hindenburg Research, etc.), and we manually collected the list of fraudulent firms from their websites. The fraudulent targets detected by these short sellers spanned the period from the start of 2010 to the end of 2023. We observed that many short-selling targets were shorted by more than one short seller,

[23] Sloan's Accrual Ratio is defined as: (Net Income - Cash Flow from Operating activities - Cash Flow from Investing activities) / Total Assets.

[24] Such as AQR Capital Management's core equity funds, DFA's growth funds, the Guggenheim defensive equity fund, the MSCI high-quality fund, and so on.

[25] Many of the well-known factors are used to the effect that any stock that ranks with relatively low factor loadings will be automatically shorted, while those that rank relatively high will be automatically longed by the multifactor hedge funds or financial institutions.

Fig. 1. Research procedure.

and most of their short-selling reports contained abundant evidence, strong logic, and solid reasoning. The fraud types included outright fraud, improper application of GAAP, fake businesses, inflated revenue, profit manipulations, hidden debt, fake assets, self-dealing, etc.

The goal of our research is to utilize training datasets to develop algorithms capable of distinguishing between short-selling targets and non-fraud samples in the test set. By training the models on known patterns of fraudulent behavior, we aim to enable the algorithms to accurately detect potential fraud in unseen data, thereby improving the identification of high-risk companies. We downloaded all short-sell targets' annual financial statement data from the "Thomson Reuters - Refinitiv Eikon" database. Fig. 3 shows the process of financial data analysis that we followed in our research.

There are 167 companies listed on the U.S. market that were reported as fraudulent by the eight short sellers mentioned, spanning from the start of 2010 to the end of 2023. Since the corporate fraud spans several years (Cecchini et al., 2010), we assumed that the shorted target companies were more or less suspicious in fraudulent activities through the years. To calculate the financial ratios accordingly, we analyzed 17 years of annual financial statement data of the 167 targets from 2007 to 2023,[26] and treated every individual year of the selected companies as a unit, namely a "company-year".

In our analysis, values that were exceptionally high or low were identified as outliers. To manage this, we adjusted the top and bottom 1 % of values for each financial feature by capping them at the 99th and 1st percentile, respectively. This approach helps mitigate the impact of extreme values, ensuring that our data remains robust and representative while avoiding distortion from anomalies. Infinity values were replaced with "NaN" (treated as missing values) and handled similarly to other missing data. We then substituted missing values with the dataset's median. To balance data exploration with quality, we implemented a straightforward preprocessing rule: any company-year sample with more than 40 % missing data was excluded. This method ensured we retained useful information while maintaining the integrity of our dataset. It is also worthwhile to mention that some shorted companies were delisted during 2010–2023. We only analyzed the company-year financial data before delisting, i.e., the company-year of the short-selling targets after the delisted year was removed.

After data cleaning, preprocessing, and labeling, we ended up with a dataset comprising 1037 short-selling company-year samples and 6305 non-fraudulent company-year samples. The ratio of fraudulent to normal company-year data in our sample is approximately 16.5 %. We split the original financial dataset into two parts: 80 % of sample data for model training and validation, which included 830 short-sell targets company-year samples, and 5044 non-fraudulent company-year samples; 20 % of the data were set for model testing, which included 207 short-selling targets company-year samples and 1261 non-fraudulent company-year samples. We then labeled all the company-year data of short-selling targets as 1, and we labeled company-year data of other firms as 0.

3.2. Features selection

Based on the literature review presented in Section 2.6, we only included the financial statement data to generate accounting ratios, binary data, and other accounting factors. We included several well-known accounting models, financial scores, and red flags (as well as their corresponding ingredients) as follows (more details can be found in Appendix B):

- Beneish's M-Score for detecting earning manipulation.
- Montier's C-Score for detecting "Cooking the Books" shenanigans.
- Dechow Fraud Score for detecting financial fraud.
- Piotroski F-Score for measuring financial strength.
- Mohanram G-Score for measuring competitive advantages.
- Altman *Z*-Score for predicting bankruptcy risks.
- Ohlson O-Score for predicting bankruptcy risks.
- "ROIC-WACC" for measuring economic value added.
- Red flags and other accounting variables.

4. Methodology

4.1. ANOVA F-test

Before running the models, we reconsidered the financial features (Fig. 2). We calculated the correlations between each pair of features and built up a "correlation matrix" based on the correlations (Fig. 4). Each small square in the matrix represents a correlation between two variables (Taylor, 1990). We can observe that there are some squares that are dark blue or dark red, representing very high or very low correlations. The possible reasons for this include: (a) Some financial metrics are calculated with similar financial statement components or in a similar way (for example, *Z-score-RTA* and *Z-Score* have a very high positive correlation), and (b) there is an accordance or a trade-off between some financial metrics.[27]

Strong positive (or negative) correlations could lead to a high multicollinearity, which can increase the complexity of the model with redundant features. It is preferable to have a sufficiently good model with fewer variables. Another concern is that some features might not be so important for the results, that is, we need to consider the statistical significance of the features. Based on these considerations, we used the ANOVA F-test to judge the importance of the different variables.[28] ANOVA, short for "Analysis of Variance", analyzes variations between and among the group means in a particular sample (Elssied et al., 2014; Dhal & Azad, 2022). Features with larger F-test values (along with smaller *p*-values) have a larger significance. Following Kumar et al. (2015), the formulae of the ANOVA F-test value are summarized as:

$$Between\ sum\ of\ squares\ (BSS) = n_1(\overline{X}_1 - \overline{X})^2 + n_2(\overline{X}_2 - \overline{X})^2 + \dots \tag{1.1}$$

$$Between\ mean\ squares\ (BMS) = BSS/df \tag{1.2}$$

$$Within\ sum\ of\ squares\ (WSS) = (n_1 - 1)\sigma_1{}^2 + (n_2 - 1)\sigma_2{}^2 + \dots \tag{1.3}$$

$$Within\ mean\ squares\ (WMS) = WSS/df_w \tag{1.4}$$

$$F = BMS/WMS \tag{1.5}$$

where df = degrees of freedom, $df_w = (N - k)$, σ_k = standard deviation of the samples in group k, N = Number of samples, k = Number of groups, and n_k = number of samples in group k.

The test statistic F is thus calculated using formula (1.5). With the idea of univariate feature selection, we aim to find out features which are statistically significant and drop the variables with relatively small F.

4.2. Evaluation metrics

Following our binary classification model construction, four possible outcomes were generated. In the cases where the activist short sellers

[26] For calculating financial metrics, for example the G-score, we selected the data from 2007, since some financial metrics require a time span of two to four years. However, the dataset of "company-year" in training and testing samples starts from 2010.

[27] For example, Current Liabilities is positively correlated with CLCA and negatively correlated with *Z*-score-NWCTA.

[28] In the Python package *scikit-learn*, we used the function "SelectKBest" to select *k* variables using the ANOVA F-test value (for classification) and selected the most significant features for our further experiments (Pedregosa et al., 2011).

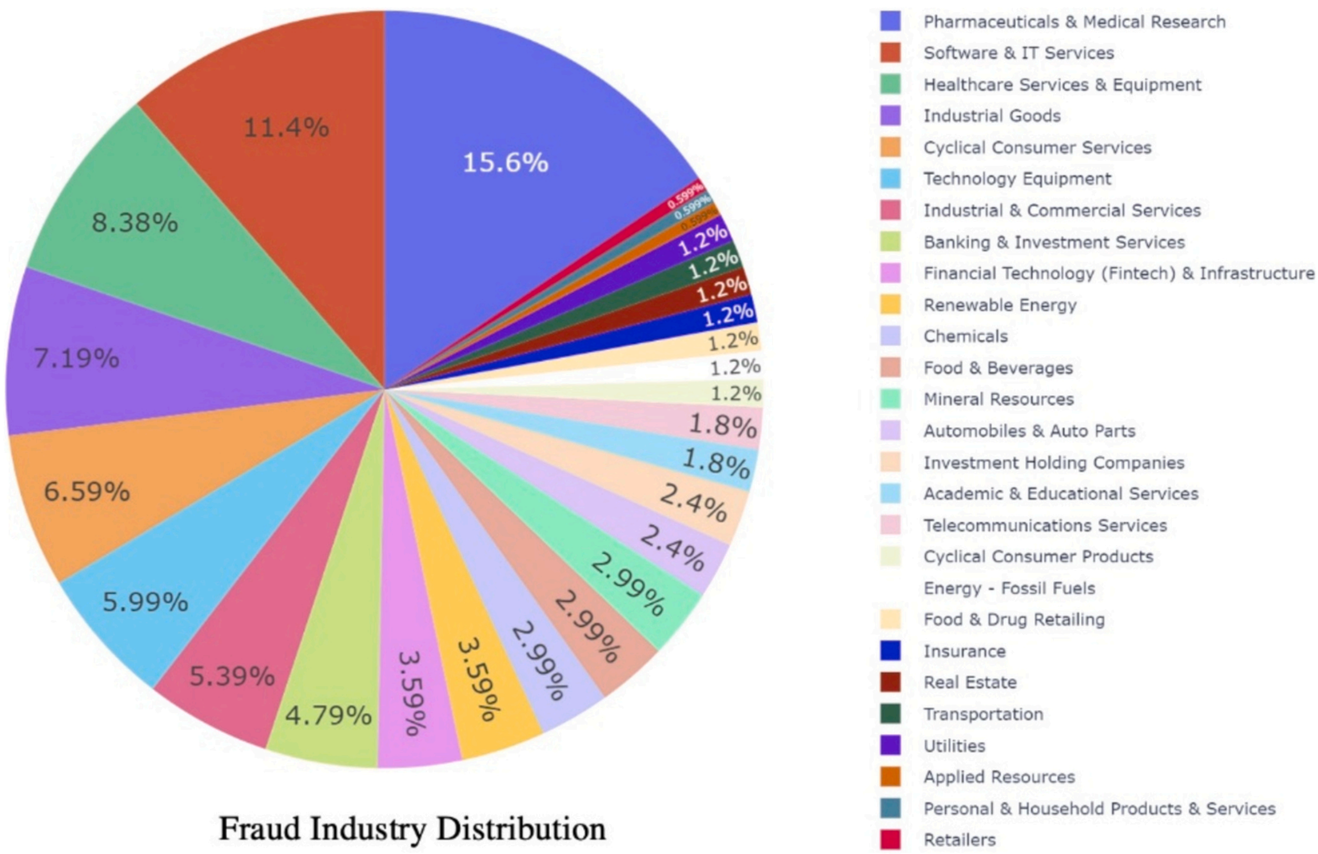

Fig. 2. MSCI GICS Industry Group distribution of 167 fraudulent companies. The fraudulent companies were defined by the short sellers, spanning from the start of 2010 to the end of 2023. We manually collected the data from the websites of the short sellers listed in the main text.

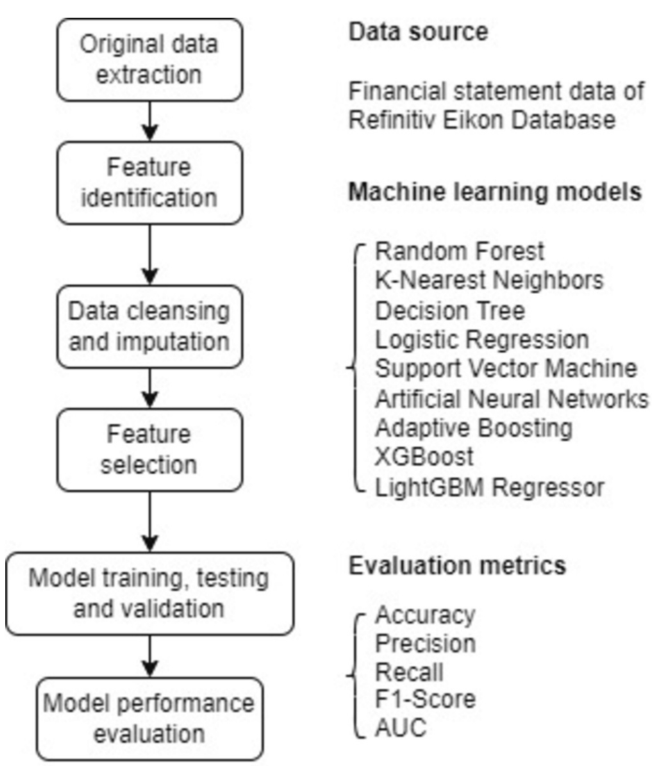

Fig. 3. Flowchart of our financial statement fraud detection approach step by step.

did not short stocks, our model could either correctly predict that they are non-fraud stocks, yielding a true negative (TN), or falsely predict that these are fraud stocks, yielding a false positive (FP). In contrast, for the cases where the activist short sellers shorted stocks, our models could either correctly classify them as a short sellers' picks, yielding a true positive (TP), or falsely classify some of them as non-fraudulent companies, yielding a false negative (FN). These four results can be categorized into a 2 × 2 contingency table, also known as a "confusion matrix". From the four outcomes, there were three evaluation metrics that we considered: "Precision", "Recall", and "F1-score". Precision thus measures the fraction of true positives among all positive samples:

$$Precision\ (P) = \frac{TP}{TP + FP} \tag{1.6}$$

Recall quantifies the classifiers' completeness:

$$Recall\ (R) = \frac{TP}{TP + FN} \tag{1.7}$$

The F1-score, also called F1-measure, is a particular case of the F_{beta}-measure (Sasaki, 2007) and is defined as follows:

$$F1\ score = \frac{2PR}{P + R} = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \tag{1.8}$$

We also included the ratio "Accuracy", defined as $\frac{TP+TN}{TP+TN+FP+FN}$, for comparing our model performance with the models in the former research, yet we did not consider it important for judging our models, because in reality, the ratio of fraudulent firms to normal firms is too small (less than 5 % detected fraud cases). With such a severely imbalanced dataset, e.g., if the model classifies all firms as non-fraud, "Accuracy" would still be around 95 %, which generates meaningless interpretations. Some of the previous research used accuracy as one of

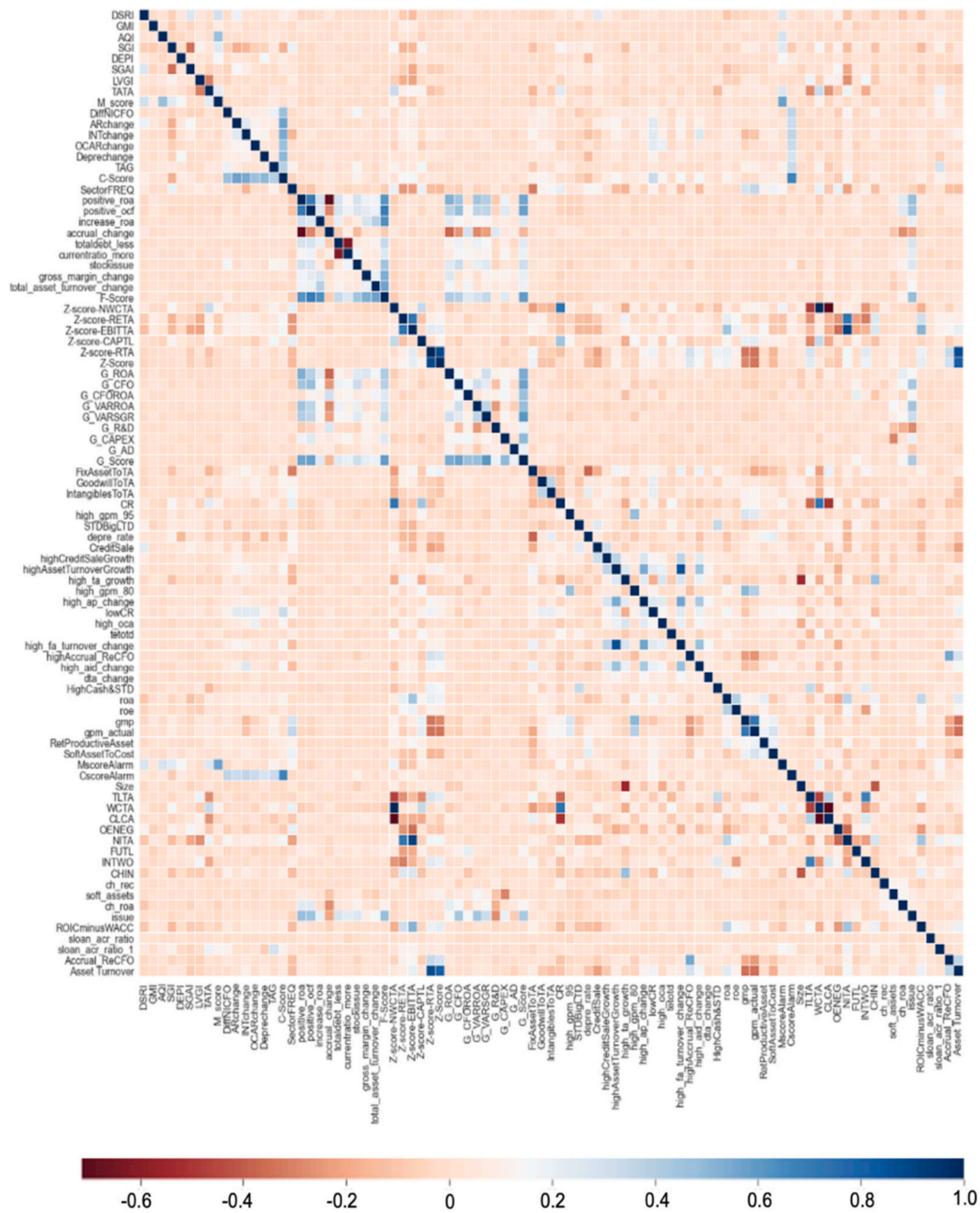

Fig. 4. Correlation matrix of selected financial features. We included all the accounting models, financial scores, red flags as well as their corresponding ingredients. The data ranges from 2010 to 2023. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the evaluation metrics, which could be misleading. The Receiver Operating Characteristics (ROC) graph illustrates the trade-off between recall and loss of specificity in the classification analysis. The discrete classifiers can be plotted as single points in a graph according to their true positive rate (TPR) and false positive rate (FPR), both of which range from 0 to 1.

Varying the decision thresholds yields a probability curve on the ROC graph, in which FPR and TPR are defined as x and y axes, respectively. According to Fawcett (2006), the ROC curve for evaluating model performance is advantageous in that the ROC curves do not change when the class distribution of the predicted targets changes.

The Area Under the Curve (AUC) is the area under the ROC curve. The AUC represents the degree or measure of separability, showing how well the model can classify the two classes. The AUC can take values between 0 and 1, and the closer the AUC is to 1, the closer the model is to a perfect classifier. In contrast, the closer the AUC is to 0, the worse the model is at separating the classes, (in fact, an AUC of 0 means that the model predicts exactly the opposite of a perfect classifier). When the AUC is 0.5, the model cannot separate the classes and is performing at the same level that a random classifier would.

In fact, misclassification costs (of false positives and false negatives) could be different for various audience. For investors and auditors, a false negative error (misclassifying fraudulent firms as non-fraudulent firms) usually costs much more than a false positive error (misclassifying non-fraud firms as fraud firms), since a false negative will lead to substantial capital losses, or potentially disastrous litigation costs (while the cost of a false positive can be priced at the cost of the human labor needed for further investigations). Thus, "Recall" and "F1-score" are the metrics that are more relevant to the goals of this paper.

4.3. Models used

4.3.1. Logistic regression

We use Logistic Regression, as its historical pervasiveness makes it a well-established baseline performance measure for all the other models we evaluate. Binary Logistic Regression solves the problem of translating Linear Regression predictions into a class likelihood:

$$P(c = 1 \mid x) = \frac{1}{1 + e^{a_0 - \sum_i a_i x_i}} \tag{1.9}$$

In our case, c is the outcome of a single sample and x is the data describing this sample. $P(c = 1| x)$ denotes the probability of the firm being involved in fraud given the data x, while $a_i, i \in 1 \ldots n$ are the underlying linear model coefficients and a_0 is its intercept. In the following steps, we denote $\sigma(x) := \frac{1}{1+e^{-x}}$ and $a^T x := \sum_{i=0}^{n} a_i * x_i$ *with* $x_0 = 1$ for ease of notation. To train our model, we additionally formulate the complement likelihood,

$$P(c = 0 \mid x) = 1 - \sigma(a^T x) \tag{1.10}$$

leading to the general expression for a single sample:

$$P(c \mid x) = \sigma(a^T x)^f \left(1 - \sigma(a^T x)\right)^{1-f} \tag{1.11}$$

where f describes the outcome, with $f = 1$ for fraudulent cases and $f = 0$ for normal cases. The expression for the likelihood for the model parameter vector a over the whole training dataset with m samples thus becomes

$$L(a) = \prod_{j=1}^{m} P\left(c = f_j | x\right) = \prod_{j=1}^{m} \sigma(a^T x_j)^{f_j} \left(1 - \sigma(a^T x_j)\right)^{1-f_j} \tag{1.12}$$

It is convenient to work with the logarithm of the likelihood

$$logL(a) = \sum_{j=1}^{m} f_j * ln(\sigma(a^T x_j)) + \left(1 - f_j\right) * ln(1 - \sigma(a^T x_j)) \tag{1.13}$$

Since we want to maximize the likelihood and, by extension, the log-likelihood, this corresponds to equating to zero for all partial derivatives with respect to the model parameters

$$\frac{\partial logL(a)}{\partial a_i} = \sum_{j=1}^{m} \left(f_j - \sigma(a^T x_j)\right) * x_{i,j} \tag{1.14}$$

As the analytical solution is hard to find, we can solve this set of equations computationally by using, for example, gradient ascent. For this, we set the initial parameters $a_{i,t=0}$ to random values and perform the updating step

$$a_{i,t+1} = a_{i,t} + \varepsilon * \frac{\partial logL(a_t)}{\partial a_{i,t}} \tag{1.15}$$

for a parameter ε, which is small enough, until some predefined stopping criteria have been met. A modern implementation of Logistic Regression can be found in *Logistic Regression for Data Mining and High-Dimensional Classification* (Komarek, 2004).

4.3.2. K-nearest neighbors

The main virtue of the K-Nearest Neighbors method is that it does not rely on any parameter, and it only needs a few (namely k) meta-parameters, relaxing the need for tight training supervision in the training and model selection process. First expounded by Altman (1992), it is based on the idea that samples that are close together in the variable space are more likely to be part of the same class. As this method does not rely on any parameter, there is no need for a training phase to optimize them, i.e., training consists of saving the training data in an appropriate data structure. For example, to predict a new sample i having the variables $x_j, j \in 1 \ldots n$ with n being the dimension of our variable-space, our model looks at the *one* neighbor (if $k = 1$) in the training set that is nearest according to a predetermined distance metric ($d : X \times X \rightarrow [0, \infty), X \in R^n$) to this sample, and returns the class of this neighbor. The distance metric d, is used as a similarity measure (Hu et al., 2016) for different data sets.[29]

However, if k is larger than one, this method might cause conflicting predictions. A simple way of resolving them is by majority vote among the k neighbors. In some cases, this leads to predictably problematic situations, for example, when the sample to be predicted is situated on the outcrop of a well-separated cluster, which leads to it being surrounded by out-of-class neighbors. When a prediction is arbitrated by simple majority vote, this example would result in a false classification. A way to solve this problem is by weighing the influence of the neighbors into consideration by a function that inversely relates to the distance d, for example $\frac{1}{d}$. As we used the *sklearn* implementation (Pedregosa et al., 2011), the execution of the method depends heavily on additional intricacies detailed in *Neighborhoods Component Analysis* (Goldberger et al., 2004).

4.3.3. Decision trees

We included Decision Trees (Quinlan, 1986, 2014) among the models we evaluated for performance, because of their long-standing relevance (survey from 2007) that continues into present-day considerations of powerful algorithms (Sarker, 2021).

A Decision Tree is constituted by a set of decisions organized in the form of a tree. These decisions consist of several steps: First, one selects a single variable from the data, then one introduces a discriminative criterion – usually by introducing a threshold into real-valued (or integer-valued) variables or associating the samples to binary-valued variables – that splits the whole dataset into two parts. Choosing a variable can be based on various criteria; and the information gain is the most widespread one in use (Quinlan, 1986). Finally, for each part of the newly segmented dataset, one can either pick a new discriminative criterion, or stop the recursion and return to a final prediction. In our case, we want to let the algorithm decide whether a company's stock should be a target of short sellers or not.

Note that one might encounter the same variable multiple times along the tree. If there were no stopping criterion to the recursion, we would end up placing every single sample in a leaf node of its own, overfitting the model to the data to the maximum degree possible. A common remedy is to specify a maximum depth to the tree, a minimum-required information gain to be allowed to further split a node, or a minimum number of samples a leaf node should contain.

Predicting the class of a single sample is achieved by traversing the trained tree from the root node, comparing the sample's value of the sample variable to be predicted against the thresholds specified by the node of the decision tree, and proceeding to its child node according to the comparison outcomes. This procedure is repeated until a leaf node is reached, and the prediction returned by the model is the one specified in this leaf node.

Using Decision Trees is very useful to determine the "variable ranking": Based on the full set, we can calculate all conditional probabilities of subsets dependent on every variable. Then, we select the variable that can provide the highest purity[30] as the first binary node of the tree. We keep repeating this process to split child groups. Based on this "recursive partitioning" process, we can conclude that the most critical variable is the one placed at the root of the tree, and the further away the children's nodes to the root, the less purity the corresponding discriminative variables have.

[29] Consider the feature vectors $A = (x_1, x_2, \ldots, x_m)$ and $B = (y_1, y_2, \ldots, y_m)$, where m is the dimensionality of the feature space. To measure the distance between A and B, one often uses the normalized Euclidean metric defined by $d(A, B) = \sqrt{\frac{\sum_{i=1}^{m} (x_i - y_i)^2}{m}}$.

[30] Purity is a concept that is based on the fraction of the data elements in the classification that belong to the subset. Purity can be defined as the frequency of its most common constituent. The variables that provide the clearest divisions of subsets will be selected as the branch nodes.

4.3.4. Random forest

As previously discussed, a drawback of Decision Trees is their susceptibility to overfitting. Instead of optimizing the tree hyperparameters, one can opt for using a Random Forest instead, which offers better guarantees regarding their generalization error (Breiman, 2001).

A Random Forest combines several Decision Trees into an ensemble. If all trees were trained using the same training data, their predictions would be highly correlated, as training a Decision Tree is a deterministic process. For this reason, a process needs to be introduced that guarantees that the single predictors – the Decision Trees – behave less correlated with each other. Modern Random Forests rely on the procedure advanced by Breiman (2001), that combines bootstrap aggregation of the training samples (drawing several training samples from the whole training set), training a single tree on this selection, with variable subsampling (at each node split, instead of considering every variable, only a randomly selected subset of variables is considered). Note that by using bootstrap aggregation, each single tree has access to its own "cross-validation" dataset. It has been shown (Breiman, 2001) that the average scores on the out-of-bag samples serve as a robust estimate of Decision Tree performance on the test dataset.

Since straightforward prediction is hampered by the generation of multiple predictions by multiple trees, a final decision is made by the majority vote over all trees in the ensemble. Deciding on variable importance cannot be done directly, but it can be done by using the out-of-bag score estimate as a proxy. For each variable, the predictions on the training set are re-run, but with that variable replaced by random noise. If this variable contributed in any way to the predictive performance of the Decision Forest, one would expect a performance degradation, were this variable randomized. By comparing the relative performance loss of all variable randomizations, an approximation to variable importance can be constructed.

4.3.5. Support vector machines

A Support Vector Machine (SVM), introduces a linear boundary in the hyper-space representation of the data, making it work similarly to the logistic regression model mentioned earlier. The data can be transformed to accommodate nonlinear class boundaries before finding an optimal boundary. These boundaries are found by constructing an optimal hyperplane separating positive from negative samples. This is done by minimizing the function

$$f = \frac{1}{2}v^T v + C\sum_{j=1}^{n}\xi_j \tag{1.16}$$

where v is a normal vector to the optimal hyperplane, and C is a freely selectable penalty weight for the slack – i.e., mathematically necessary to guarantee convergence, but not a meaningful part of the final predictive model - variables ξ_j.

with respect to the constraints

$$y_j\left(v\, x_j + b\right) \geq 1 - \xi_j \forall j \in 1\ldots n \tag{1.17}$$

where b is a constant,

$$\xi_j \geq 0 \forall j \in 1\ldots n$$

with

$$v_k = \sum_{j=1}^{n}\alpha_j{}^k y_j x_j \tag{1.18}$$

where α_j is a normal vector to the optimal hyperplane, ξ_j is a slack variable, y_j *and* x_j are elements of input vectors y (sample class) and x (sample data).

Transforming our features into an N-dimensional vector by applying the function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^N$, we transform all instances of x_i with $\varphi(x_i)$. This (feature-transformed) minimization problem has the dual quadratic optimization problem

$$\sum_{i=1}^{N}\alpha_i - \frac{1}{2}\left(\alpha^T D\alpha + \frac{\alpha_{max}^2}{C}\right) \tag{1.19}$$

with respect to the constraints

$$\alpha^T y = 0 \tag{1.20}$$

$$D_{ij} = y_i y_j K\left(x_i, x_j\right) \tag{1.21}$$

where D_{ij} is an element of matrix D, and $K\left(x_i, x_j\right)$ is a function that determines the convolution of the dot-products (Cortes & Vapnik, 1995).

While the SVM does not directly give a probability estimate, a good surrogate is the distance to the decision boundary. A common amelioration is to introduce another round of training, applying logistic regression to the output of the SVM, as e.g., described by Platt, 1999

4.3.6. Artificial neural network

A neural network consists of several layers that are made up of so-called neurons, a layout inspired by early research into how the human brain might work. These neurons are (quasi-)differentiable functions that map an m-dimensional space to a scalar, where m is the dimension of the last layer. The input of each neuron generally comprises the weighted sum of the output of the previous layer, plus a bias term. The first layer of neurons gets its inputs from the data, while the last layer outputs the corresponding decision.

Training is done by iteratively propagating the input through each successive layer (feed-forward), and then comparing the output of the final layer and the pick decision via a loss function giving an error. In the second step, this error is then differentiated with respect to the weights and biases of the previous layer, which indicates the general direction of updating. These differentiations can then be done layer-wise, going from the last to the first layer (Cross et al., 1995). Different methods to implement or approximate this gradient descent exist. As the goal is to use existing implementations of well-known algorithms efficiently, the NAdam optimizer (Dozat, 2016), as implemented in *Keras*, was used. Since the prediction target is binary, the final layer should produce a single real scalar ranging from 0 to 1, corresponding to the probability of the sample being shorted.

4.3.7. Boosted ensembles

"Boosting" works by iteratively adding a weak learner – that is, a model with relatively few parameters and a correspondingly high error rate – to the current model in combination with the intention to correct the error of the ensemble of previous weak learners. The reason for using learners with substantially high error rates is twofold: (i) To make the overall training times reasonable, the training time for a single learner has to be comparatively short; (ii) To reduce the correlations between the predictions of different learners (if the learners were more complex, the risk that their predictions are highly correlated would rise significantly, making the ensemble model less robust).

To define the learning process more formally, let m be the m-th stage of an ensemble learning process and $f_m(x)$ a weak learner, for example, ${\beta_m}^T X_{train}$ for a linear model. Instead of fitting directly to y_{test}, one fits $f_m(x)$ to $y_{m-1} - f_{m-1}(x)$ yielding y_m (Hastie et al., 2009). The first working implementation described for ensembles of this kind was AdaBoost (Adaptive Boosting), using an ensemble of decision trees of height 1 (Freund & Schapire, 1997).

Another implementation is offered by XGBoost (Chen & Guestrin, 2016), which belongs to the class of gradient boosting algorithms, whereas the residual between the last prediction and the outcome is differentiated to train the next predictor. LightGBM employs a strategy akin to that of Ke et al. (2017), with the primary distinction between the two models being the methodology used in constructing their decision trees. Whereas XGBoost constructs its decision trees on a level-wise basis, LightGBM opts for a leaf-wise approach, adding individual leaves during the building process.

4.3.8. Cross-validation and hyperparameters

While training the above-mentioned models, most of them necessitate tuning parameters to account for the differences in the way they represent reality. Adjusting the model parameters based on their performance on the training dataset is likely to lead to overfitting, especially given that the parameter spaces for many of these models are excessively large. Therefore, the training dataset is split into n folds – such that each fold contains the same proportion of positive to negative samples, and the models, with varying parameters, are trained n times on $(n-1)$ folds, while holding back the n-th fold for validation. Averaging the scores across the n folds will deliver a robust estimate of the relatively best parameter set, while greatly reducing the risk of overfitting and selection bias. See more on cross-validation in Stone (1974).

5. Empirical results and analysis

5.1. Results and analysis

We ranked all financial features according to the variations between and among the (fraudulent and normal) group-means detected by the ANOVA F-value. Then, we gradually added the number of financial features (from only one feature to all features according to the ranking) to train the nine machine learning algorithms. Fig. 5 gives the dependence of the F1-score performance as a function of the number of financial features. With more financial features added into the algorithms, the F1-score first improves rapidly and then tends to saturate, or grow slowly, beyond typically 30 features for all machine learning algorithms. Thus, we used the top 20 financial features selected by the ANOVA F-value method for further investigations. The ensemble methods (XGBoost and LightGBM) are the best machine learning algorithms among the nine methods, and both can achieve around 80 % when using the full set of all financial features. The F1-score data shows that most of the nine models can obtain a reasonably accurate classification of the fraudulent and normal firms.

Table 1 provides the scores of the nine models, which indicate that the majority of the models performed quite well with the selected financial features. The XGBoost (XGB) had an F1-score of 82 %, with Recall 72 % and Precision 94 %. LighBGM (LGB) had the second highest F1-Score, around 79 %, with 71 % Recall and Precision of 90 %.

Fig. 6 shows that among the selected models, XGBoost (XGB) and LightGBM Regressor (LGB) have the highest AUC. The confusion matrices (see Appendix C) also confirm that many of our models can tell the difference between fraudulent and normal firms well.

5.2. Features ranking list and feature of importance

Identifying the most significant features that influence the decisions made by supervised machine learning algorithms is a valuable exercise. Algorithms such as Random Forest, Decision Trees, XGBoost, and LightGBM come equipped with "feature of importance" functions. These functions are instrumental in pinpointing which financial ratios or variables play a crucial role in the model performances. By leveraging these functions, one can gain insights into the features that substantially impact the outcomes of these algorithms, facilitating more informed adjustments and improvements to the models. We selected the top 20 important features for each model, then ranked them according to their importance. Table 2 gives the list of feature importance for the models of Random Forest (RF), Decision Trees (DT), Logistic Regression (LR), XGBoost (XGB), and LightGBM Regressor (LGB).

From the feature importance ranking lists of the five algorithms, we can achieve a broad understanding of the machine learning decision rules. According to Fig. 5, XGBoost (XGB) and LightGBM Regressor (LGB) are the first- and second-best machine learning algorithms to classify the fraudulent and normal firms in our research. It is interesting that the components of Altman's Z-score, Ohlson's O-score, Beneish's M-score, and Dechow's Fraud score rank highly in all the lists, suggesting a convergence of the fraud companies' characteristics. Z-score, O-score, M-score, and Fraud-score are well-known classical leading indicators to detect financial distress or financial manipulations.

Since the XGBoost (XGB) and LightGBM Regressor (LGB) have the highest F1-score, we described the importance of each feature as determined by these two algorithms in Fig. 7. It is obvious that XGBoost and LightGBM Regressor have different weightings of the top features of greatest importance. Appendix D gives the importance of each feature as determined by Random Forest (RF), Decision Trees (DT), and Logistic Regression (LR).

The outstanding performances suggest that activist short sellers could pay special attention to the top-ranking features ranked by the features of importance in the XGBoost and the LightGBM. The financial characteristics of these financial features (top 20) are described in the following section (see Appendix B for more detailed explanations of the financial features):

- Firms under financial distress (*LVGI, LowCR, Z-score items, OENEG, TLTA)*.
- Companies facing pressure on revenue or net earnings (*G_VARSGR, CHIN, NITA*), and showing signs of earnings manipulation in income statement items (*deprechange, SGI, AQI*) and balance sheet items (*soft_assets, ch_rec, ch_inv, FixAssetToTA, IntangiblesToTA*).
- Firms that have fast asset growth as a proxy for acquisitions (*high_ta_growth*).
- Firms with inefficient assets, operation, and capital utilization (*ROA, ROE, ROIC*).
- Businesses with low operating cash flow to cover the total liability (*FUTL*).
- Companies with high accruals (*ar_rate*) and abnormal gross margins (*high_gpm, gpm*).
- Businesses in some specific industries (*FREQ*).
- Inconsistent growth in SG&A (Sales, general, and administrative expense) over sales (*SGAI*).
- Value destruction activities (*ROICminusWACC*).

6. Polytope fraud theory and unified investor-protection framework

6.1. Polytope fraud theory

Building on financial features identified by various algorithms and insights from activist short sellers' reports, we introduce the Polytope Fraud Theory (PFT). This theory extends the Fraud Diamond Theory by adding a fifth element: Accounting Anomalies. It can be expressed as:

$$\begin{aligned}\text{PFT}: \text{Fraud} = & f\,(\text{Pressure}, \text{Opportunity}, \text{Rationalization}, \\ & \text{Capability}, \text{Accounting Anomalies}).\end{aligned}$$

Our theory offers a new approach to identifying fraudulent or high-risk cases in the stock market. It suggests that these instances often share similar financial characteristics, pointing to underlying risk commonalities. We now summarized the ten accounting commandments as a checklist for financial analysts to judge whether a listed business has committed financial statement fraud to support our theory.

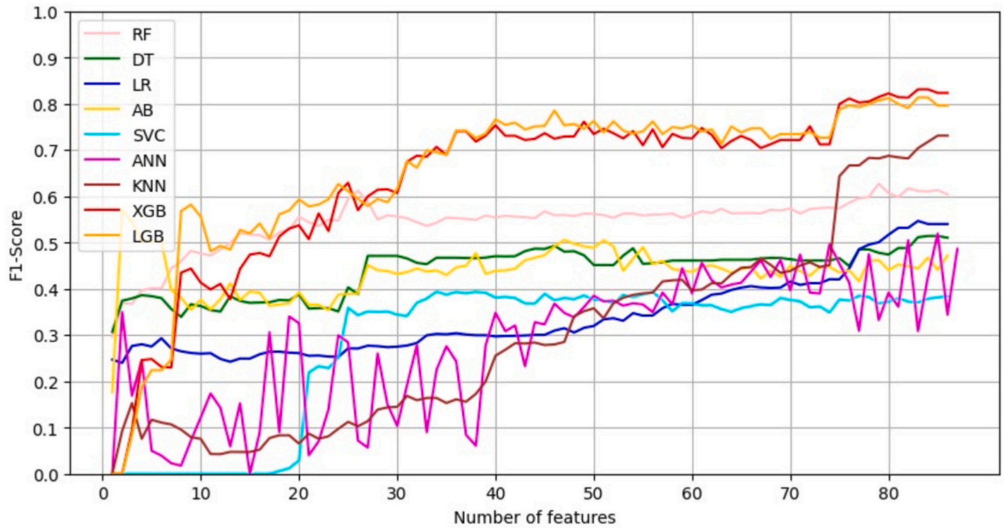

Fig. 5. Dependence of the F1-score Performance as a function of the number of features for the nine machine learning algorithms. The order of features is determined according to the ANOVA F-value.

Table 1
Scores of the nine machine learning models.[1] Higher scores means higher performances of a model. The values are rounded to two decimal places as a lower bound for the uncertainty that can be attributed to each score.

Method	Accuracy	Precision	Recall	F1 Score
XGB	0.95	0.94	0.72	0.82
LGB	0.94	0.90	0.71	0.79
AB	0.92	0.76	0.71	0.74
KNN	0.93	0.86	0.63	0.73
RF	0.86	0.55	0.74	0.63
ANN	0.80	0.42	0.81	0.56
DT	0.80	0.41	0.73	0.53
LR	0.77	0.38	0.80	0.52
SVC	0.88	0.75	0.38	0.51

[1] Accuracy = (TP + TN)/(P + N); Precision = TP/(TP + FP); Recall = TP/(TP + FN); F1-score = 2 Precision*Recall / (Precision + Recall).

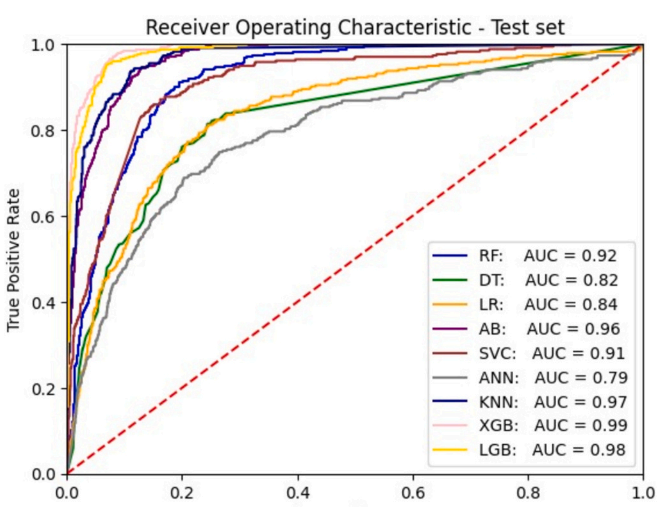

Fig. 6. ROC graph and AUC values based on the data of the test set for the nine machine learning algorithms. The curves show the ROC of each model, and the AUC scores are shown at the bottom right. ROC: receiver operating characteristics; AUC: area under the curve (of ROC).

1. **Income statement fraud bubble**: Many companies that commit financial statement fraud aim to inflate their revenue or net earnings (higher EPS) through fake income such as high accruals,[31] hidden costs or expenditures, or both. Usually, the management board of fraudulent companies intentionally fabricates sales or significantly increases credit sales, which might lead to abnormal profitability[32] compared to its industry peers. In addition, companies tend to need consistent revenue, net earnings, tax, and cash flow patterns. Sometimes, they may also have smaller dividend payout ratios[33] compared with industry peers or with their history. Investors are strongly encouraged to closely monitor high-growth companies that exhibit inconsistent revenue growth, abnormalities in receivables or payables, mismatches between revenue and tax patterns, one-time asset disposals, and improper impairment tests.
2. **Balance sheet fraud bubble**: For every fake dollar of net earnings from income, there must be somewhere to park it in the balance sheet. Thus, a long-existing fraudulent company tends to have a swollen and "filthy" balance sheet, leading to a small return on assets (ROA), or a suspicious debt structure, such as an over-reliance on short-term debt. Financial managers might use more than one balance sheet item (i.e., cash, accounts receivable, inventory, and so on) to hide fake net income produced from an income statement fraud bubble. However, the accumulation of fake net incomes will deteriorate some operating ratios and trigger many accounting irregularities.[34]
3. **Value destruction activities**: Business is for profit. A normal business is designed to make more profit than the costs incurred

[31] Due to the strict internal controls of the banking system, it is very difficult for banks to collude with fraudulent companies in committing fraud (at least in the U.S.). Thus, many fraudulent businesses record a high revenue whilst they can do little to improve real cash flow, leading to a high accrual increase or large percentage of credit sales (i.e., high accounts receivable).

[32] Fake revenue with hidden costs and expenses lead to overly high profit margins, while fake revenue and fake costs & expenses result in relatively low profit margins, compared to industry peers.

[33] If revenue is inflated and there is a lack of operating cash inflow, the discontinued dividend policies might provide some hints.

[34] There are already some alarm scores that can provide insights about the accounting irregularities in both income statements and balance sheet statements, such as the Beneish M-score, Montier's C-score, Dechow's Fraud score, etc. We also advise investors to pay attention to items with abnormal increases.

Table 2
Ranking of the Feature Importance for five different models. The feature importance is derived by the corresponding functions in the machine learning algorithms. The numbers in the left column show the ranks, based on which the machine learning algorithms classify the fraudulent and normal companies. See Appendix B for explanations of the features.

Rank	Ranking_RF	Ranking_DT	Ranking_LR	Ranking_XGB	Ranking_LGB
1	Z_Score_C2	CHIN	CHIN	OENEG	ar_rate
2	roa	OENEG	C-Score	ROICminusWACC	*Z*-score-B
3	CHIN	FREQ	FUTL	roa	FREQ
4	roe	Z_Score_C1	high_ta_growth	high_ta_growth	FixAssetToTA
5	Z_score_China	ch_rec	FREQ	rsst_acc	gmp
6	high_ta_growth	roa	ROICminusWACC	roic	Size
7	FUTL	Size	tetotd	lowCR	Z-score-C
8	ROICminusWACC	FUTL	Z-score-B	CHIN	NITA
9	NITA	ch_roa	Z_Score_C4	roe	Z-score-E
10	roic	ch_inv	F-Score	FUTL	Z-score-D
11	TLTA	Z-score-D	OENEG	G5	SGI
12	Z-score-C	ROICminusWACC	Z-score-D	Var5	TLTA
13	Z_Score_C1	Impair2	O-score	soft_assets	CHIN
14	FREQ	Z-score-C	Size	high_gpm	Z-score-A
15	SGI	lowCR	lowCR	ch_rec	DEPI
16	OENEG	Z-score-E	Var4	ch_inv	IntangiblesToTA
17	Z-score-B	WIPToTA	G3	G6	AQI
18	ar_rate	IntangiblesToTA	G5	Z-score-A	TATA
19	Z_Score_C4	C-Score	Deprechange	Z-score-D	SGAI
20	Z-score-D	Z_Score_C4	*INTchange*	ar_rate	LVGI

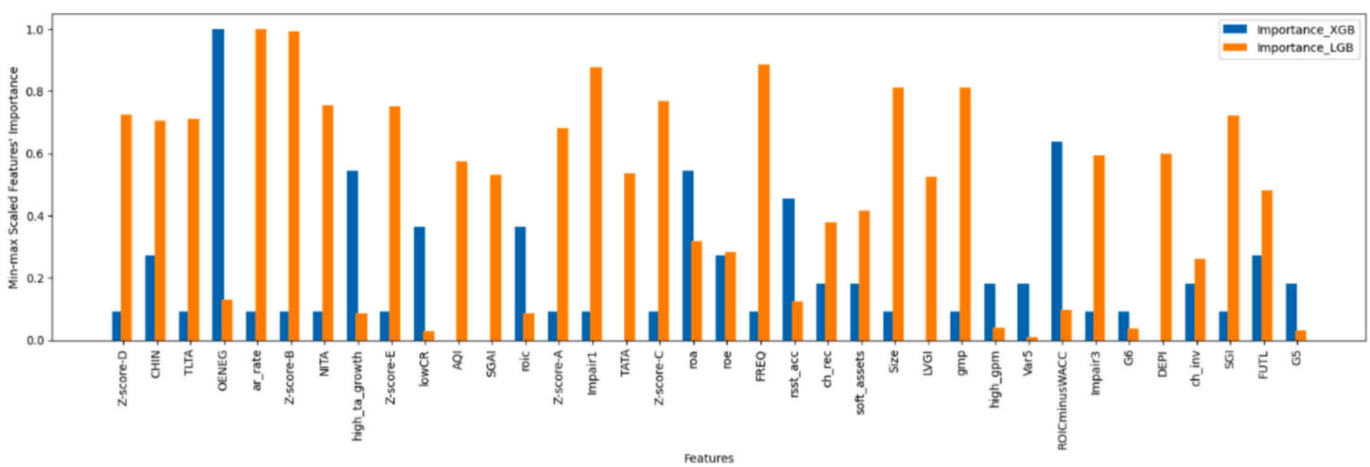

Fig. 7. Top 20 features of importance for the XGBoost and LightGBM models. The amplitude of each bar encodes the importance of the corresponding feature. The ranking of the features from left to right represents variations between and among the (fraudulent and normal) group means.

undertaking the business' activities, creating positive Economic Value-Added (EVA) ($ROIC - WACC > 0$) for its stakeholders (shareholders and creditors).Therefore, if a business consistently returns a low economic value ($ROIC - WACC < 0$), investors need to question the business model of the company.

4. **Financial distress, "Bet-on Agreement", or overreliance on financing cashflows**: Companies may falsify their financial statements to obtain gains such as stock price increases or high remunerations. In addition, companies with high leverage or low current or quick ratios tend to boost their net earnings to falsely enhance their financial statements, to obtain better credit ratings.[35] Managers of fragile businesses may only need to conduct limited fraud to significantly boost the stock price, since minimal positive signs may be enough to mislead investors. Investors should, therefore, pay special attention to companies that show signs of financial difficulties,[36] or have "Bet-on Agreements", or overreliance on financing cashflows,[37] as investing in fraudulent companies with fragile conditions can damage investor capital irreversibly if the company goes bankrupt.
5. **Suspicious transactions with related entities and self-dealing**: It is generally not easy for the company management to gain assistance internally to inflate revenue, hide costs, or create fake transactions. Thus, company insiders may create shield companies to inflate revenue or hide costs or expenses. In addition, there is a high risk that the management or controlling shareholders might conduct "self-dealing" through internal transactions. Thus, if a large percentage of transactions are between the company and its related parties, investors are advised

[35] Companies with higher credit ratings can secure funding with less effort and at reduced costs.

[36] Investors should compare a company's Altman's *Z*-score, Ohlson's O-score, interest coverage ratio, current ratio and quick ratio with other companies in the same industry to get more insights.

[37] Good companies rely on their operating cashflows. If a company's financing cashflow is much higher than its operating cashflow over multiple years, investors need to validate the cashflow against the business features.

to validate whether such transactions are detrimental to the shareholder value.

6. **Aggressive remuneration and disconcerting corporate culture**: If a company (i) experiences unrealistic revenue or expense changes compared with industry peers; or (ii) has pressure to increase earnings growth (e.g., negative net earnings or decrease in net earnings); or (iii) has aggressive remuneration for management or employees, such as huge bonuses, large company options, etc. or (iv) are with the managers displaying "unreasonable behavior" in public, investors should be aware of the corporate culture. If the company leaders express the attitude of "success at any cost" or "report only good news", or are hostile to people raising questions, the corporate culture might blind its employees or distort their ethics.
7. **Unreasonable mergers and acquisitions (M&As) and investments**: Strategic management evidence suggests that it is challenging to generate synergy from M&As, as around 83 % of mergers fail to produce any business benefit for shareholders (KPMG International Limited, 1999). When a company conducts frequent M&A activities without appropriate impairments or asset write-down, shareholders should be cautious about the company's strategic management goal. Furthermore, if the company invests in opaque projects or projects that have not been accurately capitalized, or fails to perform proper impairment tests or asset write-downs, investors should pay close attention to the quality of the assets, particularly reputation, goodwill, and other intangible assets. Revenue or net earnings growth created by external mergers and acquisitions cannot overcome weak internal operations over time.
8. **Special industries, income from offshore affiliates, and lack of regulations**: Certain industries inherently present fewer challenges for conducting fraudulent activities and pose greater difficulties for their auditing, indicating a higher likelihood of fraudulent behavior occurring. In addition, if a firm is doing business in many foreign countries, it may be very difficult for auditors to collect and validate accurate accounting data from foreign affiliations within a limited auditing timeframe. Thus, it is easier to "cook the books" for companies doing business abroad, especially in countries and regions with weak investor protection jurisdictions. Certain industries, such as fisheries and forestry, are more prone to fraudulent activities due to the inherent challenges in auditing them. Conversely, industries in periods of deregulation or rapid emergence, like financial markets in 2008 or the early cryptocurrency sector (e.g., the FTX case), often face elevated risks of fraud. These industries may lack adequate regulations to prevent malpractice, and investors may have insufficient knowledge of new markets, leading to increased opportunities for fraudulent behavior during such transitional phases.
9. **Insiders' and auditors' behavior**: If (i) the CEO or CFO resigns for no apparent reason, or (ii) auditors resign during the auditing period, or (iii) a rapid turnover occurs in the management team, or (iv) management has any history of illegal activity, or (v) some insiders sell a large number of stocks, then investors are advised to pay special attention to the upcoming financial reports. If a listed company has no internal audit department or lacks an audit committee, the company has more opportunities to commit fraud. Moreover, if the auditor does not give an unmodified opinion, or resigns after auditing the accounting report, the investor should consider such behavior as an "accounting alarm".[38] Moreover, internal management's stock-selling behavior can be considered an indicator of insider confidence.
10. **Complex corporate structure, inappropriate accounting rules, and opaque information**: If a business has a very complex group structure (e.g., with hundreds of subsidiaries or many unrelated businesses like a conglomerate), or employs inconsistent accounting rules, or provides little detail to explain important information such as tax issues, then it may be easier for the management to manipulate financial statements, because auditors have limited resources and time to validate the information.

Some of the above financial and accounting characteristics can be classified with the specific descriptions of the Fraud Diamond Theory: Characteristic 4 and 5 can be regarded as "Motivation", 6 as "Rationalization", 7, 8, and 9 can be classified as "Opportunity", and 10 can be attributed to "Capability". However, Characteristic 1, 2, and 3 cannot be allocated to any of the four elements of the Fraud Diamond Theory. Thus, we propose an additional element, namely "the Accounting Anomalies", to describe any systematic accounting irregularities appearing in balance sheets or income statements (Characteristic 1 and 2, above), and the value destruction activities described in Characteristic 3.

The Fraud Triangle Theory and the Fraud Diamond Theory describe the theoretical elements of fraud. Complementarily, our Polytope Fraud Theory (PFT) describes the practical and detailed fraudulent activities or unreasonable behavior that might occur from the corporate level. In addition, the Polytope Fraud Theory proposes a new aspect of fraud, which considers multiple "accounting alarms", or red flags, that could be found in a fraudulent company's financial statements. The following case study of one of the largest bankruptcies in U.S. history, the collapse of Enron in 2000, illustrates how the PFT identifies these common financial and accounting red flags.

6.1.1. Illustration with the case of Enron

1. Income statement fraud bubble: Enron created "hyper-inflated" revenues through two methods. First, it adopted mark-to-market (MTM) accounting for its wholesale energy contracts. The MTM accounting method treats energy contracts as financial derivative contracts, which can be used to overestimate future unrealized revenue based on hypothetical total net present value of the deal. However, MTM accounting also indicated that the market prices, which were used to value the unrealized gains or losses from of the financial derivatives as revenues, should "***reflect management's best estimate***".[39] Second, the "Enron Online" trading platform used the so-called "merchant model" of revenue, instead of the "agent model".[40] The "agent model" only records the brokerage fee earned from a deal as revenue, while the "merchant model" records the whole deal as its own revenue. In other words, Enron bought the financial derivatives and then resold them to its clients. Enron's adoption of the "merchant model" for accounting deal transactions, along with the application of mark-to-market valuations for financial derivatives, substantially inflated its reported revenues, which were largely unrealized and non-cash. In the short period between 1996 and 2000, Enron's revenue surged by more than 750 %, from 13.3 billion dollars in 1996, to $100.8 billion in 2000. In fact, from 1999 to 2000, the growth rate of revenue reached more than 150 %. Enron ranked 7th in the world in revenue in 2000 according to *Fortune Global*

[38] It is litigation risk that might lead to such unusual behavior.

[39] Emphasis added, from Page 43 of Enron's 1998 Annual Report, can be found here: https://picker.uchicago.edu/Enron/EnronAnnualReport1998.pdf

[40] The major difference is that the "agent model" does not bear the risks of the asset, while the "merchant model" does, so the latter one must take the asset rights. Financial institutions such as Goldman Sachs adopt the "agent model" since they only consider the brokerage fees of the transactions as their revenues. For example, a deal is $100, and the brokerage fee is $1. Under the "agent model", Goldman Sachs only records $1 as its revenue, while Enron records $100 as its revenue and $99 as its costs, so the net profit of Enron is $1, which is around 1 % net profit margin.

Table 3
Gross profit and net profit margin of Enron, 1996–2000 (dollars in millions).

	1996	1997	1998	1999	2000
Revenues	$13,289	$20,273	$31,260	$40,112	$100,789
Gross Profit	$2811	$2962	$4879	$5351	$6272
Gross Profit Margin	21.2 %	14.6 %	15.6 %	13.3 %	6.2 %
Net Income	$584	$105	$703	$893	$979
Net Income Margin	4.4 %	0.5 %	2.2 %	2.2 %	1.0 %

500 (source year).

Table 3 shows an opposite trend of net profit margin against rapid revenue growth from 1996 to 2000.[41] The gross margin dropped by around 15 % from 1996 to 2000, which was mainly due to the rapid increase of the wholesale segment. The wholesale segment of Enron made little income since the net profits of the wholesale segment were only some small brokerage fees. As a result, the net profit margin declined, from 4.4 % in 1996 to 1.0 % in 2000. However, Enron's management team refused to explain why the rapid revenue growths were accompanied with low gross profit margins, which raised concern of analysts.[42]

In Table 4, the whole-sale segment accounted for 94 % of the total revenue stream, while it only generated 1.8 % of the operating margin in 2000, due to the small brokerage fee mentioned above. The most profitable business was the logistics segment, including transportations and distributions, while it only accounted for 2.9 % of Enron total revenue. In addition, Enron reported $979 million dollars net income with $1.4 billion non-cash revenue in 2000. The rapid revenue growth alongside big non-cash unrealized gains led to low operating cash flows, which resulted in substantial negative free cash flow of operating activities. The Asset Turnover Ratio doubled during the same period, indicating a very fast increase in revenue over the total asset of Enron, which is suspicious. These accounting alarms show strong red flags for the Enron fraud.

2. Balance sheet fraud bubble: After years of accumulation of fake net incomes, Enron's balance sheet became very swollen. Thus, we can see deteriorations in some financial ratios. In Table 5, we can see that the Return on Assets (ROA), debt level and Return on Equity (ROE) all deteriorated from 1996 to 2000.

In Enron's balance sheets, the item "Investments in and advances to unconsolidated equity affiliates" increased from $1.7 billion dollars in 1996 to $5.3 billion in 2000, indicating a 211 % increase within four years. "Goodwill" soared from $0.87 billion dollars in 1996 to $3.6 billion in 2000. "Other Investments" surged from $1.6 billion to $5.5 billion within the same period. Moreover, compared with other major global energy competitors (Dharan & Bufkins, 2008), Enron's Return on Assets (ROA) ranked in the bottom quantile (see Table 6). Fast balance sheet expansion with high leverage and low business efficiency is clearly a red flag for Enron.[43]

3. Value destruction activities: In James Chanos' statement,[44] he

Table 4
Revenue and operating income by segments in 2000 (dollars in millions).

	Logistics	Wholesale	Retails	Broadbank	Other
Segment Revenue	$2955	$94,906	$4615	$408	($2095)
% of Total Revenue	2.9 %	94.2 %	4.6 %	0.4 %	NA
Operating Income (loss)	$565	$1668	$58	($64)	($274)
% of Total Revenue	28.9 %	85.4 %	3.0 %	−3.3 %	NA
Operating Margin*	19.1 %	1.8 %	1.3 %	−15.7 %	NA

* Operating Income as % of Segment Revenue.

Table 5
Financial ratios, 1996–2000.

	1996	1997	1998	1999	2000
Net Profit Margin	4.4 %	0.5 %	2.2 %	2.2 %	1.0 %
Asset Turnover Ratio	0.82	0.90	1.07	1.20	1.54
Return on Assets	3.6 %	0.5 %	2.4 %	2.7 %	1.5 %
Debt Ratio	76.9 %	75.1 %	76.0 %	71.3 %	82.5 %
Return on Equity	22.6 %	1.9 %	10.2 %	10.6 %	9.5 %

Table 6
Data from global energy companies (dollars in millions).

	Revenue in 2001	Profit Margin	Return on Assets
ExxonMobil	$191,581	8.0 %	10.7 %
BP	$174,218	4.6 %	5.7 %
Royal Dutch/Shell	$135,211	8.0 %	10.0 %
Enron (2000 data)	$100,789	1.0 %	0.4 %
ChevronTexaco	$99,699	3.3 %	4.2 %
Total Fina Elf	$94,312	7.3 %	9.0 %
American Elec. Power	$61,257	1.6 %	2.1 %
Duke Energy	$59,503	3.2 %	3.9 %
El Paso	$57,475	0.2 %	0.2 %
ConocoPhillips	$56,984	5.7 %	5.1 %
Petròleos de Venezuela	$46,250	7.9 %	6.0 %
Reliant Energy	$46,226	2.1 %	3.1 %
ENI	$44,637	15.5 %	6.0 %
Dynegy	$42,242	1.5 %	2.6 %

Table 7
Conservative calculations of Enron's ROIC-WACC, 1996–2000 (dollars in millions).

	1996	1997	1998	1999	2000
Operating Income	690	15	1378	802	1953
Short-term & Long-term debt	6254	6254	7357	8196	10,228
Total common shares & Minority Interest & Preferred securities	5070	7758	3103	13,000	14,788
Invested Capital (Debt & Equities)	11,324	14,012	10,460	21,196	27,194
Return on Invested Capital (ROIC*)	6.1 %	0.1 %	13.2 %	3.8 %	7.2 %
Proxy Interest rate (10 yr Bond Rate)	6.4 %	5.7 %	4.7 %	6.4 %	5.1 %
Debt to Asset Ratio	76.9 %	75.1 %	76.0 %	71.3 %	82.5 %
Cost of Capital	12 %	12 %	12 %	12 %	12 %
Weighted Average Cost of Capital (WACC)	7.7 %	7.3 %	6.5 %	8.0 %	6.3 %
ROIC-WACC	−1.6 %	−7.2 %	6.7 %	−4.2 %	0.9 %

* ROIC is calculated as Operating Income as % of Invested Capital.

discussed the reasons why he shorted Enron stocks. A major reason he gave was that Enron's cost of capital was above 7 % (at close to 9 %), while its return on invested capital was around 7 %. The negative (ROIC-WACC) indicated Enron was exhausting the invested capital. Thus, he had shorted Enron stock since November 2000. In addition, based on Enron's annual reports (1996–2000), the U.S. risk-free rates for the same

[41] Enron Annual Reports, 1997–2000, see:https://www.sec.gov/Archives/edgar/data/72859/0000072859-97-000009.txt;https://picker.uchicago.edu/Enron/EnronAnnualReport1998.pdf;https://picker.uchicago.edu/Enron/EnronAnnualReport1999.pdf;http://picker.uchicago.edu/Enron/EnronAnnualReport2000.pdf

[42] See the report, *Is Enron Overvalued?*, by Bethany McLean, Fortune Magazine: https://www.anderson.ucla.edu/documents/areas/adm/loeb/02d21.pdf.

[43] A group of students at Cornell University identified Enron's potential earnings manipulation in 1998 using Beneish's M-score, a statistical way to detect earnings manipulations. (For more information, see: https://pdfslide.net/documents/cornell-research-report-on-enron-1998.html). MacCarthy (2017) also confirmed that Enron's M-scores were abnormal in 1996, 1998, 1999, and 2000.

[44] The statement of James Chanos: https://www.sec.gov/spotlight/hedgefunds/hedge-chanos.htm.

period, and Enron's debt structure, we calculated the Weighted Average Cost of Capital (WACC)[45] and Return on Invested Capital (ROIC) as shown in Table 7. We can see that the economic value-added from Enron business were close to or less than the weighted average cost of capital of Enron, indicating that Enron had not created any reasonable economic value for the stakeholders between 1996 and 2000.

4. Financial distress, "Bet-on Agreement", or overreliance on financing cashflows: In Enron's annual report in 2000, management highlighted the debt problem, emphasizing the importance of retaining an investment-grade credit rating, since it was critical to maintaining adequate liquidity[46] (Page 27 of the 2000 annual report). On December 31, 2000, Enron had around 2112 million dollars of long-term debt matured, accounting for close to 25 % of its total long-term debt. Thus, the urgent need to refinance the matured debt at the end of 2000 provided a strong incentive for management to maintain the debt covenants, since the debt ratio in 2000 had reached a five-year high of 82.5 %. Thus, Enron management created a "Byzantine structure" to shift the debt off its balance sheet and onto its unconsolidated affiliates, i.e., SPEs (Special Purpose Entities). In addition, Enron had significant negative free cash flow from operation from 1996 to 2000 due to the non-cash revenue over the years, which means Enron had to seek substantial external capital such as equity and debt to support the operation of the business.

5. Suspicious internal transactions and self-dealing: Enron had hundreds of SPEs by 2001 (Healy & Palepu, 2003), many of which were used to fund the purchase of derivatives contracts with gas producers under long-term fixed contracts.[47] We noticed many suspicious internal transactions red flags in Enron's 2000 annual report: Enron used the equity method (Page 37) to treat the unconsolidated affiliates to avoid consolidation (shift the debt out of Enron), and the transactions between the corporation and the unconsolidated companies could be recorded as revenue for Enron. The annual report also revealed that it guaranteed (Page 48) and held exactly 50 % net voting interest[48] in many of the unconsolidated affiliates, such as Citrus, JEDI, JEDI II, SK-Enron Co. Ltd., Whitewing Associates, etc. (Page 42). Moreover, Enron had some of its key employees, including its CFO Andrew Fastow who worked as partners of those affiliates (Page 47), indicating that these SPEs were not independent at all.

6. Aggressive remuneration and disconcerting corporate culture: Enron linked the compensation of key executives to its revenue size. In proxy statements from 1997 and 2001, the pay targeting policies indicated that the pay level of the CEO and senior management should be informed by those of companies with comparable revenue size, while the bonus levels should be also informed from that of high performing companies. This indicated that the revenue size was a factor that determined the management remunerations. In addition, the proxy statement of 2001 also outlined that within 60 days of the proxy date, stock options (13 % of common shares outstanding) were granted to the senior management team without any restrictions on subsequent sale of stocks acquired. As for its employees, Enron also implemented an aggressive renumeration system designed by CEO Jeffrey Skilling.[49] Furthermore, analyst who questioned Enron were fired,[50] and anyone who asked the management team to elaborate on the business model were blocked or ignored.[51]

7. Unreasonable mergers and acquisitions (M&As) and investments: Enron conducted various mergers and acquisitions from 1996 to 2000 without disclosing related information (Savage & Miree, 2003). In Enron's balance sheet, the "Goodwill" and "Other" items increased significantly. In 2000, the combination of "Goodwill" and "Other" accounted for 14 % of Enron's total assets. In late 2001, Enron announced a series of asset write-offs: $287 million for Azure, $180 million for broadband investments, and $544 million for other investments. Later, Enron sold Portland General Corporation with a 1.1-billion-dollar loss. These buys and sells suggested that Enron had opaque and unreasonable M&A activities, and low balance sheet quality.

8. Special industries, income from offshore affiliates, and lack of regulations: To achieve further growth, Enron implemented a diversification strategy. It extended its natural gas business to the financial market, becoming a derivatives trader and market maker in the electric power, coal, steel, water, and broadband sectors. By 2001, Enron had become a global conglomerate with businesses in Central and South America, Eastern Europe, Africa, the Middle East, and India. The Maharashtra state government halted Enron's Dabhol power station project in India on 3 August 1995 due to "lack of transparency, alleged padded costs, and environmental hazards" (Mehta, 2000). Additionally, the deregulation of the energy sector in the late 1980s means Enron has a "more flexible way" to enter the newly established energy derivatives market.

9. Insiders' and auditors' behaviors: In early 2000, $924 million dollars in shares and $1.26 billion dollars in convertible debt securities were sold by company "insiders".[52] Over the next 18 months, 68 of Enron's senior executives resigned.[53] On 12 February 2001, Kenneth Lay stepped down as the CEO of Enron but remained Chairman. On 14 August 2001, Jeffery Skilling, Lay's replacement as CEO, unexpectedly resigned and sold almost 60 million dollars worth of Enron shares. On 24 October that year, CFO Andrew Fastow was replaced by Jeffrey McMahon. Furthermore, in 2000, Enron paid Arthur Andersen 25 million dollars in audit fees, accounting for 27 % of total audit fees of Arthur Andersen's Houston office. The same year Enron paid 25 million dollars in consulting fees.

10. Complex corporate structure, inappropriate accounting rules, and opaque information: Enron refused to explain the mark-to-market (MTM) method in detail in the financial statements and did not provide appropriate disclosure regarding merger and acquisition activities nor the impairment rules to test the quality of those assets. Management refused to explain the details of the price-management asset in the balance sheets, the earnings sources, and the M&As (Savage & Miree, 2003). In addition, Enron had mysterious unconsolidated affiliates and hundreds of SPE structures (Dharan & Bufkins, 2008). Thus, analysts claimed that Enron was a "black box", and a reporter from Fortune Magazine wrote an early warning report[54] in March 2001 to raise

[45] Enron's cost of equity was assumed to be 12 % (Healy and Palepu, 2003).

[46] Enron: 2000 Annual Report: http://picker.uchicago.edu/Enron/EnronAnnualReport2000.pdf.

[47] See Tufano and Bhatnagar (1994) for a detailed description of the structures.

[48] If the voting power is above 51 %, then the balance sheet of the affiliates should be consolidated to Enron according to the rule of acquisition method.

[49] The base salary at Enron was 51 % higher than its peer group, bonus payments were 383 % higher, and stock options were 484 % higher. In addition, the bottom 15 % quantile employees would be fired.

[50] On 21 August 2021, Chung Wu of UBS PaineWebber expressed concerns about Enron's financial future and advised his clients to sell Enron stocks. He was fired on the same day. Daniel Scotto of BNP Paribas lowered his recommendation on Enron stock on 23 August 2021, and was fired shortly after.

[51] During the 17 April 2002 Enron's conference call, Richard Grubman of Highfield Capital asked Jeffery Skilling, the CEO of Enron, to explain what the price-risk-management asset in the balance sheet is, Skilling did not provide any detailed information, and after the conversation, Skilling complained about Grubman in foul language with an unmuted Microphone.

[52] For more information, see https://ssqq.com/archive/cheatinginsidertrading.htm.

[53] For more information, see https://www.latimes.com/archives/la-xpm-2002-jan-26-mn-24913-story.html.

[54] See the report *Is Enron overvalued?* by Bethany McLean, Fortune Magazine: https://www.anderson.ucla.edu/documents/areas/adm/loeb/02d21.pdf.

Table 8
Investor Protection Checklist based on Polytope Fraud Theory. This checklist is designed to help detect potential fraudulent companies in practical applications. It identifies ten critical financial and managerial practices vulnerable to manipulation or may signal underlying fraud. Investors should be cautious if a company's due diligence process triggers more than three items. If a company triggers more than seven items, investors are strongly advised to avoid the investment altogether. This tool aims to streamline the due diligence process by highlighting critical areas of concern.

Checklist Item	Description	Indicator to Watch
Inconsistent Income Patterns	Monitor high-growth companies for inconsistent revenue, tax, and cash flow patterns. Watch for abnormalities in receivables/payables and one-time asset disposals.	Inconsistent revenue, net profit growth, tax mismatches, significant asset disposals, etc.
Balance Sheet Irregularities	Analyze the balance sheet for swollen assets, suspicious debt structures, and deteriorating operating ratios caused by fake net income.	Low ROA, over-reliance on short-term debt, deteriorating ratios, improper impairment tests, etc.
Negative Economic Value Added (EVA)	Evaluate whether the Economic Value Added (EVA) is consistently low or negative, indicating potential value-destructive activities.	ROIC-WACC <0
Financial Distress	Watch for signs of financial distress, such as high leverage, low liquidity ratios, or reliance on financing cash flows.	High leverage, low current/quick ratios, long history of negative free cash flow or over-reliance on financing cashflow, etc.
Related Party Transactions	Investigate transactions with related entities for potential self-dealing that could harm shareholder value.	High percentage of related-party transactions relative to revenue.
Remuneration Practices	Aggressive remuneration policies can lead to a disconcerting corporate culture that incentivizes financial manipulation.	Above industry remuneration or aggressive incentive structure.
Questionable M&A and Investment	Assess the reasonableness and transparency of M&A activities and investments, especially when they are frequent or lack a clear strategic benefit.	Frequent M&A, invest in opaque projects.
Special Industries, Offshore Affiliates, and lack of regulations	Exercise caution with companies in special industries or those generating income from offshore affiliates, especially in emerging countries with weak legal or regulatory systems. Pay special attention to emerging technologies where regulation may be insufficient.	Offshore income from emerging countries, special industry risks, emerging technology, lack of regulation, etc.
Insider and Auditor Behaviors	Monitor insider actions such as sudden resignations of high-level executives, stock sell-offs, auditor resignations, etc., as these may signal upcoming financial issues due to information asymmetry.	Executive resignations, auditor changes, significant insider selling, etc.
Complex Corporate Structures and lack of transparency	Be cautious of overly complex corporate structures, inconsistent accounting practices, or opaque disclosures, as these may facilitate financial manipulation.	Complex structures, opaque information, inconsistent accounting rules, etc.

questions about Enron's business model and valuation.

In summary, the application of the ten accounting criteria of the checklist proposed by the **Polytope Fraud Theory** would have allowed financial analysts to conclude with reasonable conviction that Enron had committed serious frauds before its collapse revealed their full extent. This is a case-in-point illustrating that many fraudulent cases share many of the same financial characteristics (see Table 8).

6.2. *Unified investor-protection framework*

Building upon the theoretical framework for earthquake prediction by Sykes et al. (1999), we aim to adapt and apply this classification system to summarize and categorize existing investor protection and fraudulent action theories. This allows us to propose a **Unified Investor Protection Framework (UIPF)**, which integrates various fraud detection and prevention strategies (see Fig. 8). The UIPF offers a structured approach to understanding how different forces contribute to financial fraud and suggests ways to enhance investor protection across sectors.

Although some of the most notorious fraud cases (Enron, WorldCom, etc.) occurred in the U.S. market, common law countries still have a generally superior investor protection system to that of civil law countries. As a plausible measure of this superiority, the average financial market capitalization to GNP ratio of the Commonwealth countries is much larger than that of civil law countries (La Porta et al., 2008).

Due to the flexibility of the common law system, common law countries have better shareholder rights protection, while civil law countries – where investors rely more on financial intermediaries such as banks to collect information and conduct financing activities, rather than direct investing – have less well-functioning financial markets (Graff, 2008). Besides, common law countries rely on the civil law system to address social problems. For instance, the U.S. SEC introduced in 2002 the Sarbanes-Oxley Act, based on the fraud triangle theory, to reinforce investor protection in the financial market, responding to failures due to financial fraud (e.g., the Enron Scandal). Chen et al. (2015) showed that the Sarbanes-Oxley Act reduces fraudulent financial reporting and increases investor protection.

No legal system is perfect, and fraudulent cases are always occurring in financial markets. We surmise that the existence of activist short sellers complements the legal system for investor protection owing to their use of sophisticated accounting expertise to detect fraudulent listed firms. In essence, their pursuit of fraudulent companies, acting like "Batman in Gotham City", brings greater scrutiny to firms engaged in financial misconduct.

Based on the **Polytope Fraud Theory (PFT)**, we have developed an

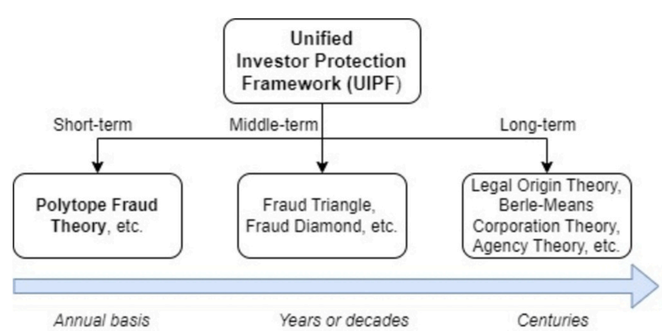

Fig. 8. Investor protection theories regarding financial fraud can be categorized into three levels. Micro-level theories focus on detecting fraud through detailed accounting and operational warning signals observed in daily practices. Meso-level theories pertain to financial criminology, abstracting common financial traits and patterns of fraudulent behavior. Macro-level theories relate to classical finance law and corporate governance principles, which have shaped financial markets for centuries, establishing broad frameworks for fraud prevention and regulation. This layered approach aids in understanding fraud from granular details to overarching systemic governance.

Investor Protection Checklist for practical application. This checklist helps investors conduct thorough due diligence by identifying key red flags associated with potential fraud. During the due diligence process, investors should give careful attention to the items listed, which highlight critical financial and operational signals that could indicate fraudulent behavior. This practical tool offers a structured approach to enhance investor vigilance and reduce exposure to fraudulent schemes.

7. Conclusion

We have investigated the performance of various machine learning methods in detecting financial statement fraud using published information. This work is the first to label companies shorted by activist short sellers (that were manually collected) and train algorithms to classify these fraudulent cases against normal cases. This paper is also the first to combine well-known asset pricing factors (mainly used by passive short sellers such as hedge funds) with accounting red flags (from accounting expertise) in financial feature selections.

In terms of performance, we found that ensemble methods such as XGBoost and LightGBM algorithms outperformed the SVM, ANN, Logistic Regression, and other machine learning models. The F1-score reaches 82 % and 79 % respectively for XGBoost and LightGBM, while for the Logistic Regression, KNN, DT, Random Forest, SVM, ANN, and AdaBoost, the F1-score ranged between 50 % and 70 %.

From a technical standpoint, our approach offers several key advantages. First, we address the issue of missing data more effectively than previous research, which often excluded valuable short-selling target companies entirely. Instead, we established a threshold for data selection: company-year data were removed only if more than 40 % of the financial features were missing. For the remaining data, missing values were replaced with the dataset median to retain as many financial features as possible. Additionally, XGBoost and LightGBM Regressor handle missing data efficiently by adjusting the weights accordingly, allowing us to preserve valuable information even with minor gaps in the data. Second, through detailed reverse engineering, our algorithms successfully identified patterns in fraud firms targeted by sophisticated activist short sellers. We attribute the strong performance of our methods to the depth of accounting insights, forensic analysis, and sharp business acumen of these short sellers. The results suggest that leading activist short sellers apply consistent, methodical fraud detection techniques, selecting short-selling targets based on solid accounting rationale.

Our findings suggest that automating the advanced financial statement analysis methods employed by short sellers is not only feasible but can significantly improve auditing processes and enhance investment strategies. This approach helps to avoid "torpedo stocks" in portfolio management while identifying potential short-selling targets. Additionally, the list of key financial features we identified—such as bankruptcy risk, industry clustering, inconsistent profitability, high accruals, and questionable business operations—indicates that fraud-prone companies share common traits. These align well with established fraud theories, further validating our approach.

Our research highlights are as follows. (1) We introduced the Polytope Fraud Theory (PFT), which outlines ten key financial red flags that may indicate a fraudulent firm. (2) We developed the Unified Investor Protection Framework (UIPF), which categorizes investor protection and fraud detection theories at macro, *meso*, and micro levels. The UIPF also provides a practical Investor Protection Checklist for real-world application, helping investors better detect and mitigate fraud in financial markets.

8. Suggestions for future research

1. Unstructured Data: Since we only included features which are closely related to the financial statement structure data (mainly accounting models, ratios, and quality factors), future research can utilize Natural Language Processing (NLP) technique in analyzing the unstructured data such as annual report footnotes, management outlook and news. Additionally, alternative data (e.g., market sentiments) and valuation data can also be considered in further research.
2. Fraud detection and accounting standards: Only U.S. fraud companies shorted by global activist short sellers were investigated. Our labeled targets follow the U.S. GAAP. Researchers may also add short-selling targets from Australia, the U.K., Germany, Hong Kong, Switzerland and other countries and regions that follow different accounting principles and standards (e.g., IFRS) to develop more comprehensive and universal models.
3. Algorithm design and feature engineering: In this research, we focused on supervised learning algorithms, yet future studies could explore other machine learning approaches, such as unsupervised or semi-supervised methods. Additionally, while we selected 89 financial features, including engineered features from prior work, further advancements in feature engineering could significantly enhance model performance. Researchers may experiment with different algorithms, incorporate novel engineered features, and test models on diverse sample datasets to improve validation and robustness.

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Appendix A. Description of the financial investigating institutions and their websites, where we manually collected the fraud company list mentioned in section 3.1

Muddy Waters Research, a private-owned hedge fund, is based in the U.S. and actively conducts research while also taking positions that are in line with their research (mainly short-selling targets). It tries to identify fraudulent publicly listed firms in Asia, Europe, and North America by using forensic accountants, trained investigators, and valuation experts to carry out due diligence reports, which are released online to the public. The CEO of Muddy Waters Research is Carson Block.

Website: https://www.muddywatersresearch.com/research/.

GMT Research is a Hong Kong-based accounting research firm. It conducts deep fundamental accounting research on global listed firms and issues reports to uncover financial anomalies and accounting misbehavior. It issues reports to subscribing financial institutions (i.e., hedge funds and long-only investors) and does not take positions on the companies it investigates. The company is made up of a group of forensic experts led by Gillem Tulloch.

Website: https://www.gmtresearch.com/en/about-us/hall-of-shame/?offset=0&limit=20.

Citron Research, a privately-owned American hedge fund, publishes online reports on listed firms that are engaged in fraud. The reports are produced by a team of forensic investigators led by Andrew Left. The company mainly focuses on the U.S. market, and not only holds long side investments, but also short sells fraudulent firms before it issues reports on its website.

Website: https://www.citronresearch.com/.

Hindenburg Research, an American investment research firm, focuses on activist short-selling research, specializing in forensic financial research. It releases public reports online targeting fraud, accounting irregularities, illegal/unethical businesses, undisclosed internal transactions, etc. The company was founded by Nate Anderson.

Website: https://hindenburgresearch.com/.

Gotham City Research, a diligence-based American investing firm, focuses on companies which seem to have systematic patterns of fraud, and issues a series of reports to uncover hidden wrong-doing. Based on its publication it builds long or short equity positions. Gotham City Research was founded by Dan Yu, a previous hedge fund trader.

Website: https://www.gothamcityresearch.com/main.

J Capital Research, a U.S. investment advisory company, publishes research reports on public companies. While J Capital Research has expertise in Chinese markets, it also locates short-selling targets throughout the world. The company relies on its deep, independent, and quality primary research and publishes free reports to small, unaffiliated investors. The company is led by Anne Stevenson-Yang and Tim Murray.

Website: https://www.jcapitalresearch.com/reports.html.

MOX Reports, a private short seller's website, focuses on reverse engineering fraudulent activities such as problematic capital structure and outright fraud. Its approach is to hunt down potential fraud and explain "exogenous elements" that drive irrational price distortions against the corporate fundamentals. Richard Pearson, an ex-investment banker and founder of the website, builds long or short positions based on his research reports published on the website.

Website: https://moxreports.com/research-topics/

Glaucus Research, a U.S. investment firm, conducts investigative research on listed companies in Asia Pacific, Europe, and North America. Its team has expertise in forensic accounting, law, and investigative reporting. The firm hold long or short positions based on their research. Recently the company was separated into Blue Orca Capital, led by Soren Aandah, and Bonitas Research, led by Matthew Wiechert. Both companies focus on short selling listed fraudulent firms as per Glaucus Research's previous activities.

Websites: https://www.blueorcacapital.com/short-activism; https://www.bonitasresearch.com/research/

Appendix B. Formulae used for financial and accounting features

Table A1
Beneish's M-Score for detecting earning manipulations.

Feature	Type	Formula
DSRI	Ratio	$\frac{Receivables_t/Sales_t}{Receivables_{t-1}/Sales_{t-1}}$
GMI	Ratio	$\frac{(Sales_{t-1} - Cost\ of\ goods\ sold_{t-1})/Sales_{t-1}}{(Sales_t - Cost\ of\ goods\ sold_t)/Sales_t}$
AQI	Ratio	$\frac{(1 - Current\ Assets_t + PP\&E_t)/Total\ Assets_t}{(1 - Current\ Assets_{t-1} + PP\&E_{t-1})/Total\ Assets_{t-1}}$
SGI	Ratio	$\frac{Sales_t}{Sales_{t-1}}$
DEPI	Ratio	$\frac{Depreciation_{t-1}/(Depreciation_{t-1} + PP\&E_{t-1})}{Depreciation_t/(Depreciation_t + PP\&E_t)}$
SGAI	Ratio	$\frac{Sales, general, and\ administrative\ expense_t/Sales_t}{Sales, general, and\ administrative\ expense_{t-1}/Sales_{t-1}}$
LVGI	Ratio	$\frac{(Long\ Term\ Debt_t + Current\ Liabilities_t)/Total\ Assets_t}{(Long\ Term\ Debt_{t-1} + Current\ Liabilities_{t-1})/Total\ Assets_{t-1}}$
TATA	Ratio	$\frac{\Delta Current\ Assets_t - \Delta Cash_t - \Delta Current\ Liabilities_t - \Delta Current\ Maturities\ of\ LTD_t - \Delta Income\ tax\ payable_t - Depreciation\ and\ Amortization_t}{Total\ Assets_t}$
M_score	Ratio	$-4.84 + 0.92\ DSRI + 0.528\ GMI + 0.404\ AQI + 0.892\ SGI + 0.115\ DEPI - 0.172\ SGAI + 4.679\ TATA - 0.327 LVGI$

Table A2
Montier's C-Score for investigating "cooking the books" activities.

Feature	Type	Formula
DiffNICFO	Binary	1 if there is a growing difference between *net income and cash flow from operations*, 0 otherwise
ARchange	Binary	1 if *Days of Account Receivable* is increasing, 0 otherwise
INTchange	Binary	1 if *Days of Inventory* is growing, 0 otherwise
OCARchange	Binary	1 if *other current assets to revenues* is increasing, 0 otherwise
Deprechange	Binary	1 if there are declines in *depreciation* relative to *gross property plant and equipment*, 0 otherwise
TAG	Binary	1 if there is a high *total asset growth*, 0 otherwise
C-Score	Number	$DiffNICFO + RD + ID + OCARchange + Deprechange + TAG$

Table A3
Piotroski's F-Score for measuring fundamental strengths.

Feature	Type	Formula
positive_roa	Binary	1 if *net income before extraordinary items is* positive in the current year, 0 otherwise
positive_ocf	Binary	1 if cash flow from operations is positive, 0 otherwise
increase_roa	Binary	1 if *net income before extraordinary items* of the current year less the one of the prior years is positive, 0 otherwise
accrual_change	Binary	1 if *cash flow from operations* is larger than *net income before extraordinary items*, 0 otherwise
totaldebt_less	Binary	1 if the firm's *leverage ratio* fell in the year preceding portfolio formation, 0 otherwise
currentratio_more	Binary	1 if the historical change in the firm's *current ratio* between the current and prior year is positive, 0 otherwise
stockissue	Binary	1 if the firm did not issue common equity in the year preceding portfolio formation, 0 otherwise
gross_margin_change	Binary	1 if the firm's current year's *gross margin ratio (gross margin scaled by total sales)* less the prior year's *gross margin ratio* is positive, 0 otherwise
total_asset_turnover_change	Binary	1 if the firm's current year *asset turnover ratio* less the prior year's *asset turnover ratio* is positive, 0 otherwise
F-Score	Number	$positive\_roa + positive\_ocf + increase\_roa + accrual\_change + totaldebt\_less + currentratio\_more + stockissue + gross\_margin\_change + total\_asset\_turnover\_change$

Table A4
Altman's *Z*-Score for predicting bankruptcy risk.

Feature	Type	Formula
Z-score-NWCTA	Ratio	$Working\ Capital_t / Total\ Assets_t$
Z-score-RETA	Ratio	$Retained\ Earnings_t / Total\ Assets_t$
Z-score-EBITTA	Ratio	$Earnings\ Before\ Interest\ and\ Tax_t / Total\ Assets_t$
Z-score-CAPTL	Ratio	$Market\ Value\ of\ Equity_t / Total\ Debt_t$
Z-score-RTA	Ratio	$Revenue_t / Total\ Assets_t$
Z-Score	Ratio	$0.012\ ZscoreNWCTA + 0.014\ ZscoreRETA + 0.033\ ZscoreEBITTA + 0.006\ ZscoreCAPTL + 0.999\ ZscoreRTA$

Table A5
Mohanram's G-Score for judging competitive advantage.

Feature	Type	Formula
G_ROA	Binary	1 if $ROA_t \geq$ Industry Median ROA_t, 0 otherwise
G_CFO	Binary	1 if $\frac{Cash\ Flow\ from\ Operation_t}{Average\ Total\ Asset} \geq$ Industry Median $\frac{Cash\ Flow\ from\ Operation_t}{Average\ Total\ Asset}$, 0 otherwise
G_CFOROA	Binary	1 if $\frac{Cash\ Flow\ from\ Operation_t}{Average\ Total\ Asset} \geq ROA_t$, 0 otherwise
G_VARROA	Binary	1 if *ROA variance*$_t$ *of* 4 *years*$\leq$ Industry Median *ROA variance* of 4 years, 0 otherwise
G_VARSGR	Binary	1 if *Revenue Growth variance* of 4 years$\leq$ Industry Median *Revenue Growth variance of* 4 *years*$_t$, 0 otherwise
G_R&D	Binary	1 if $R\&D_t / Total\ Assets_t \geq$ Industry Median $R\&D_t / Total\ Assets_t$, 0 otherwise
G_CAPEX	Binary	1 if $CAPPEX_t / Total\ Assets_t \geq$ Industry Median $CAPEX_t / Total\ Assets_t$, 0 otherwise
G_AD	Binary	1 if $Adverting_t / Total\ Assets_t \geq$ Industry Median $Adverting_t / Total\ Assets_t$, 0 otherwise
G_Score	Number	Sum of *G1 - G8*

Table A6
Ohlson's O-Score for predicting bankruptcy risk.

Feature	Type	Formula
Size	Ratio	$log\left(\frac{Total\ Assets_t}{GNP_t}\right)$
TLTA	Ratio	$\frac{Total\ Liabilities_t}{Total\ Assets_t}$
WCTA	Ratio	$\frac{Working\ Capital_t}{Total\ Assets_t}$
CLCA	Ratio	$\frac{Current\ Liabilities_t}{Current\ Assets_t}$
OENEG	Ratio	1 if $Total\ Liabilities_t > Total\ Assets_t$, 0 otherwise
NITA	Ratio	$\frac{Net\ Income_t}{Total\ Assets_t}$
FUTL	Ratio	$\frac{Funds\ provided\ by\ Operations_t}{Total\ Liabilities_t}$
INTWO	Ratio	1 if net income was negative for the last two years, 0 otherwise
CHIN	Ratio	$\frac{Net\ Income_t - Net\ Income_{t-1}}{\lvert Net\ Income_t \rvert + \lvert Net\ Income_{t-1} \rvert}$
O-score	Ratio	$-1.32 - 0.407\ Size + 6.03\ TLTA - 1.43\ WCTA + 0.0757\ CLCA - 1.72\ OENEG - 2.37\ NITA - 1.83\ FUTL + 0.285\ INTWO - 0.521\ CHIN$

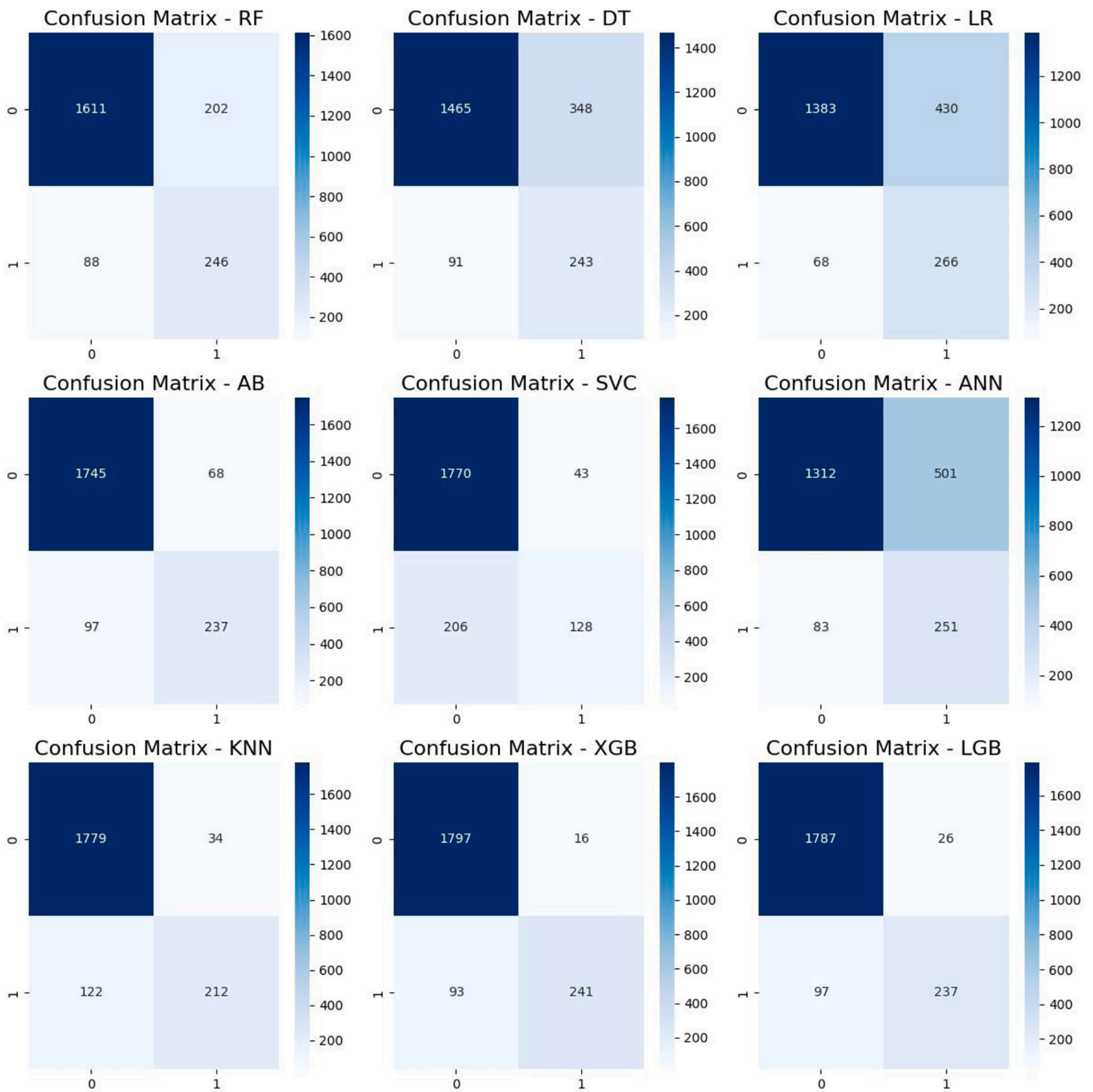
Table A7
Dechow's Fraud-score for judging financial fraud.

Feature	Type	Formula
rsst_acc	Ratio	$change\ in\left(\frac{Shareholder\ Equity - Pre\ Stock - Cash\&equivalents}{Average\ Total\ Asset}\right)$
ch_rec	Ratio	$\frac{Change\ in\ Receivable}{Average\ Total\ Asset}$
ch_inv	Ratio	$\frac{Change\ in\ Inventories}{Average\ Total\ Asset}$
soft_assets	Ratio	$\frac{Total\ Assets - net\ PP\&E - Cash\&equivalents}{Average\ Total\ Asset}$
ch_cs	Ratio	$Change\ in\ (Revenue - change\ in\ Receivables)$
ch_roa	Ratio	$change\ in\left(\frac{Net\ Income}{Average\ Total\ Asset}\right)$
issue	Binary	1 if company issued long-term debt or common and/or preferred equity, 0 otherwise

Table A8
Red flags and financial variables.

Feature	Type	Formula
SectorFREQ	Ratio	The frequency of the firm's industry appears in the list of shorted companies over the total fraudulent samples.
FixAssetToTA	Ratio	$Total\ Property, Plant\ And\ Equipment_t / Total\ Assets_t$
GoodwillToTA	Ratio	$Goodwill\ Actual\ Value_t / Total\ Assets_t$
IntangiblesToTA	Ratio	$Intangibles\ Gross_t / Total\ Assets_t$
WIPToTA	Ratio	$Construction\ in\ Progress / Total\ Assets_t$
STDBigLTD	Binary	1 if $Short\ Term\ Debt_t / Current\ Liabilities_t > 0.75$, 0 otherwise
depre_rate	Ratio	$Depreciation_t / (Depreciation_t + Fixed\ Asset_t)$
CreditSale	Ratio	$Net\ Accounts\ Receivable_t / Revenue_t$
HighCash&STD	Binary	1 if $Short\ Term\ Debt_t / Total\ Asset_t > 0.15$ and $Cash\&equivalent_t / Total\ Assets_t > 0.15$, 0 otherwise
highAssetTurnover Growth	Binary	1 if the absolute value of *Asset Turnover change ratio* > 0.1, 0 otherwise
high_ta_growth	Binary	1 if *Total Asset Growth* > 0.1, 0 otherwise
high_gpm	Binary	1 if the growth of *Gross Profit Margin* is in top 20 % of the industry, 0 otherwise
Accrual_ReCFO	Ratio	$Revenue_t / Cash\ Flow\ from\ Operation_t$
highAccrual_ReCFO	Binary	1 if $Revenue_t / Cash\ Flow\ from\ Operation_t$ is in top 20 % of the industry, 0 otherwise
high_ap_change	Binary	1 if the absolute change of $Accounts\ Payable_t / Revenue_t > 0.1$, 0 otherwise
lowCR	Binary	1 if $Current\ Ratio_t < 1$, 0 otherwise
highCreditSaleGrowth	Binary	1 if $Accounts\ Receivable_t / Revenue_t > 0.1$, 0 otherwise
high_oca	Binary	1 if the change of $Other\ Current\ Assets_t > 0.1$, 0 otherwise
tetotd	Ratio	$Total\ Equity_t / Total\ Debt_t$
Asset_Turnover	Ratio	$Revenue_t / Total\ Asset_t$
high_aid_change	Binary	1 if the change *of Inventory Days* > 0.1, 0 otherwise
dta_change	Binary	1 if the change of *Deferred Tax Assets* > 0.1, 0 otherwise
MscoreAlarm	Binary	1 if *M_score* exceeds -1.78, 0 otherwise
roa	Ratio	$Return\ On\ Assets_t$
roe	Ratio	$Return\ On\ Equity_t$
CscoreAlarm	Binary	1 if *C-Score* exceeds 2, 0 otherwise
gmp	Ratio	$Gross\ Margin_t$
RetProductiveAsset	Binary	1 if $Operating\ profit_t / (PP\&E_t + Inventory_t)$ is in the top 10 % quintile, 0 otherwise
SoftAssetToCost	Binary	1 if $(Total\ assets_t - PP\&E_t + Inventory_t) / Cost\ Of\ Goods\ Sold_t$ in the top 10 % quintile, 0 otherwise
roic	Ratio	$Return\ On\ Invested\ Capital_t$
ROICminusWACC	Ratio	$Return\ On\ Invested\ Capital_t - Weighted\ Average\ Cost\ of\ Capital_t$
Sloan_accruals	Ratio	$(change\ in\ Current\ Assets - change\ in\ Cash) - (change\ in\ Current\ Liability - change\ in\ Short\ Term\ Dedt - change\ in\ Tax\ Pabable) - depreciation$
Sloan_acr_ratio	Ratio	$\left(\frac{Income\ from\ Operation_t - Sloan\_accruals}{Average\ Total\ Asset}\right)$
Sloan_acr_ratio_1	Ratio	$\left(\frac{Net\ Income_t - Cash\ Flow\ from\ Operation_t - Cash\ Flow\ from\ Investment_t}{Total\ Asset_t}\right)$

Appendix C. Confusion matrices derived by each model



Note: For each confusion matrix, each matrix entry stands respectively for: True Negative (top left), False Positive (top right), False Negative (bottom left) and True Positive (bottom right). The number in each box stands for the number of data records in the test set.

Data availability

Data will be made available on request.

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